

GENERAL THETA FOUNDATIONS I: ADAPTIVE TESTING AND COLLISION-SENSITIVE COMPARISON

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ABSTRACT. We study comparison of marked causal experiments with fixed persistent state and separately priced hidden or visible randomness. On the nonconvex observable behavior image we construct an independent adaptive testing game and polynomial test classes with attained finite-level duality. In behavior dimension a and diameter R , their error is at most $4Ra/\sqrt{n}$, using at most $(n+1)^a$ occupied test coefficients; this does not charge redundant simulator rows. We then construct a two-sphere scattering experiment with an actual correlated mark and a gate chosen after the first noisy report. Its private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector gives error below $13/100$, at the same width-one register. Removing the collision makes every deficiency zero. Exact finite values, uniform collision bounds and computable rational presentations prove this separation without convexifying the private image. Finally, marked variation controls the entire optional-projection process; Gaussian observation of graph-core approximants yields a quantitative microscopic changing-filtration limit. The costed comparison, strict-certificate, task and domain theory is developed with its separate information and analytic hypotheses.

CONTENTS

1. Introduction	2
1.1. Main results	3
1.2. Relation to classical theory	4
1.3. Organization and analytic scope	4
2. Intrinsic causal deficiency	4
2.1. Experiments and executable signatures	4
2.2. The finite comparison theorem	6
2.3. A delayed channel and a strict signature separation	8
2.4. Optimal resource sets and decision certificates	9
3. Adaptive testing on the observable behavior image	10
3.1. The testing game	10
3.2. A size bound in behavior dimension	11
3.3. A three-weight certificate invisible to constant tests	13
3.4. Strict certificates and effective physical input	14
4. A marked resource gap generated by a binary collision	14
4.1. The exact marked two-report experiment	14
4.2. A full-dimensional collision preparation	15
4.3. Uniformly effective finite presentations	17
5. Uniform stability of optional projections	18
5.1. A microscopic sufficient condition	19
5.2. Local testing at the prescribed private budget	20

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5.3. Transferring intrinsic certificates across presentations	23
5.4. Witness size in the intrinsic behavior dimension	23
6. Polynomial testing of the original resource image	25
6.1. A sharp two-time nonconvex example	27
7. Endogenous common decisions	27
7.1. A shared encoder on a controlled tree	27
7.2. Lossy common design and intrinsic comparison	29
7.3. Task resolution, positive error, and the complete state	30
8. Regenerative average risks	32
8.1. The reset signature	32
8.2. Cycle comparison before optimization	33
9. An active microscopic realization	34
9.1. Interventions and the unbounded flow	34
9.2. A graph-core approximation and controlled words	35
9.3. Actual marked acquisition under feedback	36
9.4. Active regenerative average decisions	37
9.5. A posteriori bounds from graph and intervention residuals	38
10. Finite rational presentations of Gaussian acquisitions	40
10.1. A stateless report channel	40
10.2. Rationalization without a small-prefix denominator	41
11. Enriched defect certificates and their convergence	43
12. Effective graph data from regular collision charts	45
13. Physical intrinsic resource certification	48
14. Certified data for active hard spheres	49
14.1. A conserved mode with an exact marked deficiency	49
14.2. A nonzero collision-compatible graph residual	51
15. A nonconserved collision-compatible acquisition	52
15.1. One actual mark, one nonzero residual, and a positive certificate	54
16. A joint physical and nonconvex certificate theorem	56
16.1. Exponential acquisition values	57
Appendix A. Theorem dependencies and information conventions	58
References	58

1. INTRODUCTION

Comparison of experiments asks whether the information supplied by one observation system can replace that supplied by another. A causal comparison must also respect the order in which observations and actions become available. A resource-constrained comparison asks a further question: how much persistent state and shared randomness does the replacement require? These questions differ even for two binary reports. A simulator that cannot retain or reread its first emission cannot in general reproduce a correlated second emission. Revealing a persistent random selector to the tester is also different from hiding it. The distinction concerns the joint laws being compared, not merely the implementation of a convex combination.

We study marked experiments. A mark is an actual latent quantity of the source or target system, not a fresh variable sampled after a comparison. The comparison tests the joint law of the mark and the controlled report history. Private simulation uses finite retained registers and fresh private randomization. A hidden selector is an additional priced persistent variable; a visible selector is exposed and must occur with the same independent reference law on the target side.

With these conventions the attainable behavior image is compact but usually nonconvex. The central issue is to compare this original image rather than a convex relaxation with a different information pattern.

1.1. Main results. For fixed finite instruments and a resource signature, let B denote the observable discrepancy image and $F(b) = \max_i f_i(b)$ its marked feedback error. The f_i are affine event tests, including zero. The deficiency is $\delta = \min_B F$. An adaptive test is an independently chosen function $\kappa : B \rightarrow \Delta_m$, scored at b by $\sum_i \kappa_i(b) f_i(b)$. It observes a proposed law for purposes of verification; it is not an added physical information channel.

Theorems 3.1 and 3.3 give the independent variational identity

$$\delta = \sup_{\kappa} \min_{b \in B} \sum_i \kappa_i(b) f_i(b) = \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa} \int \sum_i \kappa_i(b) f_i(b) d\mu(b),$$

and a quantitative polynomial hierarchy. In a bounded chart of the observable affine space, of dimension a and metric diameter R , its attained finite-level values satisfy

$$0 \leq \delta - d_n \leq 4Ra/\sqrt{n}.$$

An approximating test needs at most $(n+1)^a$ occupied Bernstein coefficients, and the corresponding moment dual needs at most $(n+2)^a$ support points. The dimension counts behavior, not row variables. The nonlinear score prevents replacement of a lottery over revealed candidate laws by its barycenter. At degree zero such a replacement is possible, and the resulting convex relaxation can miss a strictly positive private deficiency.

Proposition 3.5 gives a concrete nonlinear witness with three cubic behavior weights, nine nonnegative derivative terms and seventeen positive rational coefficients. It certifies a private lower bound above $2/5$, while constant tests give zero.

The independent game and its size theorem complement the strict polynomial certificate hierarchy proved in Theorem 6.1. The latter remains an exact verifier on a chosen row cube; it is not itself called an independent dual. Strict lower levels are complete, and their verification complexity is kept separate from the behavior-coordinate degree bound. With uniformly effective stateless finite presentations this produces certified intervals at the original state and randomization budgets.

The second main result makes collision mechanics essential. We prepare two spheres in full-dimensional incoming phase-space boxes and observe a bounded function of the tagged transverse velocity after scattering. An actual binary mark, correlated with the incoming impact orientation, is retained but never revealed. A binary gate is chosen after the first noisy report. Theorem 4.1 first solves the limiting finite experiment exactly:

$$\delta^P = \delta_r^v = \sqrt{5/2} - 9/8, \quad \delta_2^h = 1/8 \quad \text{at width } (1, 1).$$

Uniform nongrazing collision estimates and Gaussian tails then give, for contact distance $9/10 \leq d \leq 11/10$ and noise $0 < \sigma \leq 1/12$,

$$\delta^P, \delta_r^v > 9/20, \quad \delta_2^h < 13/100.$$

The same detector and preparation have zero deficiency for $0 < d \leq 2/5$, when no collision reaches the detector during acquisition. Thus conserved collective motion cannot account for the gap. The estimates are uniform on an interval of contact distances and under geometric similarity. Explicit collision charts also produce rational marked presentations with certified errors tending to zero.

The third result strengthens the conditional-law content of marked comparison. For equivalent joint laws with a common latent marginal, Theorem 5.1 constructs a latent-preserving coupling under which the squared uniform distance between bounded posterior martingales is at most 84

times the joint total variation. The coupling is a convergence device, not an executable simulator. Gaussian observation of an invariant microscopic flow gives

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \|h_n - h\|_{L^2(\mu)},$$

so graph-core approximation yields convergence of entire optional projection processes. The maximum-over-time estimate, rather than a finite-dimensional argument, supplies the changing-filtration input. Finite-checkpoint risk transport still uses the stateless identity comparison and keeps its original budget.

1.2. Relation to classical theory. Classical statistical comparison and randomization originate with Blackwell and Le Cam [1, 2, 13]. Filtered comparison has important antecedents, including Norberg [3]. We do not claim priority for introducing causal deficiency. In the adapted setting Weisshaupt’s Definition 5.1 and Theorems 5.1–5.2 [19] use a compact convex closure of adapted stochastic operators, allowing finitely additive outputs. A fixed finite-state private image need not have that convexity. The two-report examples identify the precise failed step in inserting the state constraint into such a separation proof. Our testing game retains the nonconvex image by changing the verifier’s strategy space, not by asserting a new unrestricted randomization criterion. Its minimax ingredient is classical [18].

Bernstein positivity is likewise not new here. Theorem 1.4 of de Klerk–Laurent [20] gives constructive hypercube degree and error bounds through Bernstein and Handelman representations. Laurent–Slot [21], Theorem 2 and Corollary 3, obtain stronger Schmüdgen degree bounds using Jackson kernels. Theorem 9 and Proposition 2 of de Klerk–Vera [22] connect Chebyshev coefficient norms to effective hypercube certificates. Our bound concerns an independent simplex-valued adaptive test on a nonconvex behavior image; it is not an improvement to those general positivity bounds. Pulling that test back to row coordinates still incurs a separate positivity-verification cost.

The finite-state, common-information and automata antecedents [5, 4, 12, 17] remain relevant to the task and synthesis theory. The physical flow and collision rule are classical [6]. The contribution of the scattering construction is the marked, feedback-sensitive resource comparison with uniform quantitative certificates in that flow, not a new mechanics of elastic collision.

1.3. Organization and analytic scope. After the exact comparison conventions, Sections 3 and 4 prove the adaptive hierarchy and its collision-sensitive realization. Section 5 proves the posterior-process estimate and its microscopic sufficient condition. The remaining sections develop the complete finite certificate, task, regeneration, graph-domain and physical-presentation machinery used for broader acquisitions, including the explicitly solvable collective benchmark. Appendix A records the theorem dependencies in mathematical terms.

All physical conclusions have their stated finite-time and preparation hypotheses. A binary scattering chart and a fixed-particle optional-projection estimate are different from particle-uniform recollision estimates or a nonlinear kinetic limit. In particular a large-deviation action, a hierarchy corrector uniform on action sublevels, and a limiting nonlinear resolvent require their own analytic inputs. The comparison theorem transports consequences of such inputs; it does not create them by naming a dependency.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual

mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m &\in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' &\in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) *For compatible experiments E, F, G and profiles K, L ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product

with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1 - \varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual

terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

3.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 3.1 (Independent adaptive testing duality). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{3.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting

test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

3.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 3.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (3.2)$$

For $n \geq 0$ let $\beta_{\alpha, n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha, n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha, n}(x) \ell_i(x). \quad (3.3)$$

Theorem 3.3 (Intrinsic adaptive degree and support bounds). *The maxima in (3.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (3.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (3.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (3.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (3.5) cannot be replaced by its behavior barycenter. For $n=0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 6.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 3.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the

visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

3.3. A three-weight certificate invisible to constant tests.

Proposition 3.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (3.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (3.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

3.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 3.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

4. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

4.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (4.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 4.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^p(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (4.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

4.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (4.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (4.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 4.2 (Uniform transverse collision signal). *For every preparation in (4.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (4.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4, 113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (4.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 4.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^v(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (4.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which

tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 4.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

4.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 4.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 3.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for

example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (4.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

5. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 5.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (5.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (5.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (5.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy | z) - Q(dy | z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (5.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (5.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 5.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) | \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) | \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 5.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

5.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 5.3 (Collision-observation optional-projection limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (5.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (5.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (5.3). Applying Theorem 5.1 proves (5.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (5.3). That is the resource-comparison consequence. The latent-preserving coupling in (5.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

5.2. Local testing at the prescribed private budget. The convex converse in Theorem 2.2 has an explicit selector price. We now give a different converse that leaves the private register unchanged. Its witnesses are finite covers of the policy space by local testing inequalities, rather than one separating functional on a convex hull. Throughout this subsection the experiments, horizon, and alphabets are finite; for effective statements their instrument probabilities are rational. The prescribed widths and, when present, selector cardinality are fixed.

Let $\mathcal{P}$ be the product of the row simplexes of the private transducer. For a hidden or visible selector with at most r values, include its simplex and r copies of the transducer rows in $\mathcal{P}$. Zero selector weights are permitted. The relevant deficiency has the form

$$\delta = \min_{q \in \mathcal{P}} F(q), \quad F(q) = \max_{i \in I} f_i(q), \quad 0 \leq F \leq 1. \quad (5.5)$$

Here I consists of all parameter, feedback-tree, and marked-event tests. The empty event is included. For a visible selector a test specifies a feedback tree and event in each selector fiber. Thus

$$f_i(\lambda, q) = \sum_u \lambda_u \{Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)\}.$$

For a hidden selector the target term is independent of λ . Every f_i is affine in each row separately, with rational coefficients. This follows by expanding complete controlled paths: a time-indexed row is used at most once along each path. It holds simultaneously for all feedback tests, not just for open-loop words.

For an integer $M \geq 1$, intersect a row simplex Δ_{d-1} with each closed coordinate box

$$\prod_{j=1}^d [b_j/M, (b_j + 1)/M], \quad b_j \in \{0, \dots, M-1\}.$$

Discard empty intersections. Products of these row polytopes form a finite closed cover $\mathcal{C}_M$ of $\mathcal{P}$. Overlapping faces cause no ambiguity. Write $V(C)$ for the products of row vertices of a cell C . These are exactly its vertices. They are rational and are obtained by setting all but at most one coordinate of each row to a box endpoint and imposing that the coordinates sum to one. This prescription also includes lower-dimensional cells and one-point rows. Define

$$\ell(C) = \max_{i \in I} \min_{v \in V(C)} f_i(v), \quad u(C) = \min_{v \in V(C)} F(v), \quad (5.6)$$

$$L_M = \min_{C \in \mathcal{C}_M} \ell(C), \quad U_M = \min_{C \in \mathcal{C}_M} u(C). \quad (5.7)$$

For time t , let d_t be the largest row alphabet cardinality, including both emitted symbol and next retained label. Put

$$\rho_t(M) = \begin{cases} 0, & d_t = 1, \\ \min\{1, d_t/(2M)\}, & d_t > 1, \end{cases} \quad b_M = 1 - \prod_{t=0}^T (1 - \rho_t(M)).$$

For a selector row set $s_M = 0$ if $r = 1$, and $s_M = \min\{1, r/(2M)\}$ otherwise. The three bounds used below are

$$\omega_M^p = b_M, \quad \omega_M^h = s_M + (1 - s_M)b_M, \quad \omega_M^v = \min\{1, 2s_M + (1 - s_M)b_M\}. \quad (5.8)$$

They do not count the number of histories or the number of policy rows. The visible-selector bound includes a second selector term because the reference joint experiment changes when its selector law changes.

Theorem 5.4 (Complete local testing certificates). *For any of the three fixed resource signatures in (5.5), the rational quantities (5.7) satisfy*

$$0 \leq L_M \leq \delta \leq U_M \leq 1, \quad U_M - L_M \leq \omega_M. \quad (5.9)$$

The upper witness is an executable rational transducer at the prescribed widths and selector cardinality. A lower witness consists of one test i_C for each cell and its inequalities $f_{i_C}(v) \geq L_M$ at all $v \in V(C)$.

For $M = 2^n$, the lower bounds increase and the upper bounds decrease to δ . In particular,

$$\delta = \sup_n \max_{(i_C) C \in \mathcal{C}_{2^n}} \min_{C \in \mathcal{C}_{2^n}} \min_{v \in V(C)} f_{i_C}(v) = \inf_n U_{2^n}. \quad (5.10)$$

If $\delta > \varepsilon$, a finite local testing certificate eventually proves this strict inequality. If $\delta < \varepsilon$, a rational original-budget simulator eventually proves feasibility below ε . For rational ε , non-strict feasibility, including equality, is decidable by real quantifier elimination. The optimum is algebraic, and an optimal row table can be chosen with real algebraic entries.

Proof. Express each row of $q \in C$ as a convex combination of its row vertices. Expand a separately affine f_i in those combinations, one row at a time. The resulting weights on $V(C)$ are nonnegative and sum to one. Consequently

$$\min_{q \in C} f_i(q) = \min_{v \in V(C)} f_i(v). \quad (5.11)$$

It follows that $\ell(C) \leq \min_C F \leq u(C)$ and hence $L_M \leq \delta \leq U_M$. No assertion that F itself is separately affine has been made. This distinction is essential at a randomized optimum. The empty event gives $\ell(C) \geq 0$.

We next prove the oscillation estimate used for the gap. Two probability rows in the same M -cell differ in total variation by at most $\rho_t(M)$. Run the two transducers against the same tester and source. As long as all earlier emitted symbols and retained labels coincide, the controls and source laws agree. Maximally couple the next joint output–state row. The conditional chance of a mismatch is at most $\rho_t(M)$. Thus the chance of any mismatch through time T is at most b_M . On the event of no mismatch the source can be coupled identically, including its actual mark. This bounds the difference of the probability of every marked feedback event by b_M , uniformly in the parameter and tester. A finite causal instrument provides precisely the conditional kernels needed for this coupling.

For a hidden selector first maximally couple its two laws. The selector mismatch has probability at most s_M ; on matched fibers the preceding bound is b_M . The difference of any fixed test is at most $s_M + (1 - s_M)b_M$. There is only one executed fiber, so this estimate has no extra factor r . For a visible selector write its fixed test as the expectation of $a_u(q) = Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q^u}(B_u)$. These numbers lie in $[-1, 1]$. On mismatched selector fibers their difference is at most two; on matched fibers it is at most b_M . Thus its oscillation is at most $2s_M + (1 - s_M)b_M$. This estimate does not forget the selector in either compared experiment.

Choose a vertex v_* minimizing F on $V(C)$ and an active test i_* there. For every other vertex,

$$f_{i_*}(v) \geq F(v_*) - \omega'_M = u(C) - \omega'_M,$$

where ω'_M is the untruncated corresponding oscillation bound. It follows that $u(C) - \ell(C) \leq \omega'_M$. Since both bounds lie in $[0, 1]$, truncation by one is valid. Choose a cell attaining L_M to obtain $U_M - L_M \leq \omega_M$. A vertex attaining U_M is an original row table; it is not a mixture of different designs.

Dyadic refinement only replaces each cell by subsets. By (5.11), its descendants cannot lower the minimum of a fixed test and hence cannot lower L_M . Every coarse vertex belongs to a finer cell and remains an extreme point of that subset; therefore the set of available vertices can only increase, and U_M cannot increase. The bounds in (5.8) tend to zero. Independent choices of the maximizing test in each of the finitely many cells give (5.10) and the two strict-decision assertions.

Finally $q \in \mathcal{P}$ and $f_i(q) \leq \varepsilon$ are finitely many polynomial conditions over the rationals. Quantifier elimination over real closed fields decides their existential conjunction; this is the classical effective Tarski–Seidenberg theorem, not a new elimination algorithm [11, §2.1, Theorem 2.1]. The set of feasible objective values is a semialgebraic ray with included endpoint δ . Its endpoint is algebraic over $\mathbb{Q}$. The nonempty optimum fiber has a point over the real algebraic numbers by the same real-closed-field transfer, proving the final assertion. $\square$

Corollary 5.5 (Private feasibility is locally test complete). *At fixed private widths, $\delta_K^p(E, F) \leq \varepsilon$ if and only if no finite dyadic local testing certificate has value greater than ε . For any finite prescribed list of resource profiles, all strictly feasible and strictly infeasible profiles can be separated by rational original-budget witnesses and finite local certificates. With rational tolerance, exact feasibility and the minimal elements in that finite list are decidable. These assertions require neither a free public selector nor a convex private behavior set.*

Proof. The first claim is (5.10). Apply the strict parts of Theorem 5.4 to each profile in the finite list and its quantifier-elimination clause to profiles on the boundary. The partial order is coordinatewise state and selector cost. Enumeration of feasible profiles then determines the minimal elements. It need not produce a single least profile, and no polynomial running time is asserted. $\square$

For the three-symbol channel V_* in Proposition 2.4, $\delta_2^{\text{mix}} = 0$ whereas $\delta_2^{\text{P}} \geq 1/(4\sqrt{3}) > 1/8$. Theorem 5.4 therefore eventually gives a finite local certificate exceeding $1/8$ at private width two. A single convex testing separator cannot have that conclusion, since the hidden convex hull already contains the target. This example identifies the logical work done by localization; it does not claim a polynomial bound on the size of the certificate.

The exact use of row vertices and real algebraic decision procedures is classical. So is the unrestricted filtered randomization criterion: The precise filtered antecedent is [10, Lemma 1 and Theorem 5], discussed in the introduction. The conclusion here is instead a complete system of local tests with executable witnesses at that fixed register, together with an explicit feedback modulus. These are different conclusions even on the finite delayed-channel slice. The original-source audit records separately the still-unverified comparison with Norberg’s full proof and with the original incomplete-machine minimization papers.

5.3. Transferring intrinsic certificates across presentations. For compatible marked experiments write $E \simeq_a \tilde{E}$ when two stateless, seed-free, marked causal channels, one in each direction, have feedback error at most a . Their report timing and action interfaces are the same. On a visible signature they compare the complete joint interface with the independent selector. These are assumptions about executable channels, not about unseeded marginal distributions.

Theorem 5.6 (Original-budget stability of intrinsic certificates). *Suppose $E \simeq_a \tilde{E}$ and $F \simeq_b \tilde{F}$. For private, fixed hidden-selector, and fixed visible-selector deficiencies at the same budget,*

$$|\delta_K(E, F) - \delta_K(\tilde{E}, \tilde{F})| \leq a + b. \quad (5.12)$$

If the approximating experiments are finite and rational, their local certificates consequently give the rigorous interval

$$\max\{0, L_M - a - b\} \leq \delta_K(E, F) \leq \min\{1, U_M + a + b\}. \quad (5.13)$$

An upper witness consists of the rational transducer and the two stateless bridge channels and has the original persistent resource budget. A lower certificate excludes every simulator at that budget for the original experiments whenever $L_M > a + b + \varepsilon$.

Proof. Compose $E \rightarrow \tilde{E}$, a near-optimal $\tilde{E} \rightarrow \tilde{F}$ transducer, and $\tilde{F} \rightarrow F$. The optimized composition inequality in Theorem 2.2, or its identical channel proof on measurable carriers, adds the three errors and multiplies the widths by $1, K, 1$. The bridges introduce no selector, so a prescribed hidden or visible cardinality remains unchanged. Reverse the bridges for the opposite inequality and take the infima. Equation (5.13) follows from Theorem 5.4. For the last assertion any original simulator would, after reverse bridging, contradict the finite lower certificate. Thus the lower statement does not assume the existence of an optimal simulator on a nonfinite carrier. $\square$

5.4. Witness size in the intrinsic behavior dimension. The local theorem can be complemented by a bound on how many different tests are needed. This bound uses the dimension of attainable behaviors, not the number of stochastic-row coordinates in a chosen realization. It does not confuse the number of tests with the cost of verifying that their violation regions cover a nonconvex set.

Let $\mathcal{B}_K = \{b(q) : q \in \mathcal{K}_K\}$ be the private behavior image from Theorem 2.2. Differences of behaviors have zero mass in each parameter–feedback row. On their linear span put

$$\|b - b'\|_* = \max_{\theta, g} \frac{1}{2} \sum_{\omega \in \Omega_g} |b_{\theta, g}(\omega) - b'_{\theta, g}(\omega)|.$$

This is a norm, $\text{diam } \mathcal{B}_K \leq 1$, and every event discrepancy $f_i(b) = Q_{\theta, g}(B) - b_{\theta, g}(B)$ is 1-Lipschitz. Let $a = \dim \text{aff } \mathcal{B}_K$; redundant realization coordinates do not change a .

Theorem 5.7 (Intrinsic-dimension testing witnesses). *Suppose $\delta_K^p(E, F) \geq L + \gamma$ with $\gamma > 0$. There is a subfamily $\mathcal{I}_0$ of the original marked feedback tests, of size at most*

$$|\mathcal{I}_0| \leq \lceil (1 + 4/\gamma)^a \rceil, \quad (5.14)$$

such that

$$\min_{q \in \mathcal{K}_K} \max_{i \in \mathcal{I}_0} f_i(b(q)) \geq L + \gamma/2. \quad (5.15)$$

Thus the same-budget private lower certificate can use this fixed subfamily on every policy cell. With rational finite input, rational L, γ , and classical real quantifier elimination, such a subfamily can be extracted effectively. No convex mixture of simulators, state increase or selector is introduced.

Proof. In the a -dimensional affine behavior space choose a maximal set $b_1, \dots, b_s \in \mathcal{B}_K$ with pairwise distances strictly greater than $r = \gamma/2$. It is an r -net by maximality. Balls of radius $r/2$ about these points have disjoint interiors and lie in a ball of radius $1 + r/2$ about b_1 . Lebesgue volume in this affine space, and its homogeneity under dilation, give $s \leq (1 + 2/r)^a$. For $a = 0$ one point suffices. At each center choose a test i_j attaining its maximal discrepancy. This discrepancy is at least $L + \gamma$. Every behavior within r of that center therefore satisfies $f_{i_j} \geq L + \gamma/2$. Take the union of these tests and remove repetitions. This proves both assertions.

For effective extraction use rational row tables as centers. Their behavior images are dense, since rational points are dense in products of simplexes and b is continuous. Given finitely many centers, the assertion that some feasible q lies at distance strictly greater than r from all centers is a finite existential semialgebraic assertion: absolute values and row maxima can be expressed by finitely many polynomial inequalities and disjunctions. Real quantifier elimination decides it [11]. When true, rational row enumeration finds such a point because the strict inequality defines a relatively open nonempty subset of the row simplexes. When false, the chosen points form an r -net. The volume bound forces termination. All discrepancies at rational centers are rational and can be compared exactly.

Finally apply Theorem 5.4 to the reduced finite family. Its maximal discrepancy has the strict gap in (5.15); sufficiently fine policy cells certify L using only these tests. The extraction bound is on the number of distinct tests, not on the number of cells or the complexity of real quantifier elimination. $\square$

Proposition 5.8 (Dimension dependence cannot be discarded). *For the delayed identity experiment on $n \geq 2$ symbols at private width one, $a = n - 1$. If*

$$0 < \gamma < \frac{1}{n(n-1)}, \quad L = 1 - \frac{1}{n} - \gamma,$$

any subfamily of original event tests that proves $\min_q \max_i f_i(q) > L$ contains at least $n = a + 1$ tests.

Proof. A private one-state behavior is a probability vector p independent of the parameter. A positive discrepancy in parameter row i has form $1 - p(B)$ with $i \in B$, and is bounded above by $1 - p_i$. If fewer than n tests are used, let S be the parameter indices of the positive-discrepancy tests. It has size at most $n - 1$. When it is nonempty choose p uniform on S . Every chosen positive discrepancy is at most $1 - 1/|S| \leq 1 - 1/(n - 1) < L$; tests with target probability zero

are nonpositive. If S is empty all tests are nonpositive. Hence no such subfamily certifies the stated bound. The full behavior set is the $(n - 1)$ -simplex, giving its dimension. $\square$

The volumetric packing step and real-algebraic elimination are classical. The conclusion here is a structural bound for the nonconvex original resource image, invariant under the addition of unobservable row coordinates. It supplies an alternative to retaining every possible event at every policy cell. It is neither a polynomial-time algorithm nor a claim that one constant convex separator characterizes the private feasible set.

6. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 3.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 6.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (6.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (6.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy–Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (6.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (6.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 6.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 6.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (6.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E}\binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (6.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 6.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (6.2) whose identity is checked by (6.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 6.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 6.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 3.3; its verification is a different cost.

6.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0,0)$ or $(1,1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 6.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 6.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

7. ENDOGENOUS COMMON DECISIONS

7.1. A shared encoder on a controlled tree. Let $\mathcal{H}_t$ be the finite supported observation–control histories of a known controlled experiment. The initial history has distribution μ , positive on $\mathcal{H}_0$. For $h \in \mathcal{H}_t$ and $a \in A_t$, write $p_t(y \mid h, a)$ for the next-report law and hay for the successor when this probability is positive. Zero-probability successors impose no condition. The finite task set is D . Task d has conditional expected stage costs $c_{t,d}(h, a)$ and terminal cost $c_{T,d}(h)$. These may be obtained by conditioning bounded losses of the actual hidden mark; the task does not alter the physical transition law. Controls do alter that law through a .

The common source-consuming encoder is

$$q_0(m \mid h), \quad q_{t+1}(m' \mid m, a, y), \quad m \in [S_t], \quad m' \in [S_{t+1}].$$

Controller d chooses its action by $\alpha_{t,d}(a \mid m)$. It cannot query the discarded history. The clock is available. The encoder does not know d ; the update sees the actual selected action, as allowed by the interface. All kernels can be stochastic. The same encoder must serve every task, even though different tasks induce different laws on histories.

Define the full-history Bellman functions and regrets by

$$\begin{aligned} V_{T,d}(h) &= c_{T,d}(h), \\ V_{t,d}(h) &= \min_{a \in A_t} \left(c_{t,d}(h, a) + \sum_y p_t(y \mid h, a) V_{t+1,d}(hay) \right), \\ G_{t,d}(h, a) &= c_{t,d}(h, a) + \sum_y p_t(y \mid h, a) V_{t+1,d}(hay) - V_{t,d}(h) \geq 0. \end{aligned} \quad (7.1)$$

The benchmark for task d is $v_d = \sum_h \mu(h) V_{0,d}(h)$. It is an ordinary full-history control value, not a posterior quantity conditioned on a fixed open-loop preparation.

Lemma 7.1 (Occupation regret). *For every admissible common encoder and task controller, the task excess satisfies the exact identity*

$$R_d - v_d = \mathbb{E}_d \sum_{t=0}^{T-1} G_{t,d}(H_t, A_t). \quad (7.2)$$

Moreover the entire vector of risks of a randomized finite-horizon design is a convex combination of risk vectors of deterministic designs with the same widths and one common encoder.

Proof. Take conditional expectations in (7.1) along the actual controlled path. Summing the resulting expressions for $G_{t,d}$ telescopes the terms $V_{t,d}(H_t)$, leaving the sum of stage and terminal costs minus $\mathbb{E} V_{0,d}(H_0)$. This proves (7.2) without assuming the control is open loop.

Sample in advance a deterministic outcome for every time-indexed row of the common encoder and every time-indexed row of every task controller. Row choices are independent with their prescribed probabilities; the sampled encoder table is shared across tasks. Along each trajectory each time-indexed row is encountered at most once. The product of its row probabilities is therefore the probability of that path under the sampled-table mixture. This identity holds for all tasks simultaneously after summing paths and costs. An additional independent visible or hidden seed is first conditioned on and then integrated. The conclusion is a mixture identity; the mixing variable is not claimed to be executable privately without retaining its index. $\square$

Definition 7.2 (Endogenous compatible cover). A cover of profile S consists of an initial map $i : \mathcal{H}_0 \rightarrow [S_0]$, common maps $\delta_t : [S_t] \times A_t \times Y_{t+1} \rightarrow [S_{t+1}]$, task action maps $a_{t,d} : [S_t] \rightarrow A_t$, and cells $C_{t,m}^d \subseteq \mathcal{H}_t$. Empty cells and overlapping cells are allowed. The requirements are

$$\begin{aligned} h &\in C_{0,i(h)}^d & (h \in \mathcal{H}_0, d \in D), \\ G_{t,d}(h, a_{t,d}(m)) &= 0 & (h \in C_{t,m}^d), \\ h \in C_{t,m}^d, p_t(y \mid h, a_{t,d}(m)) > 0 &\implies ha_{t,d}(m)y \in C_{t+1,\delta_t(m,a_{t,d}(m),y)}^d. \end{aligned} \quad (7.3)$$

Thus the cells cover the subtree generated by the selected actions. They need not cover histories generated by actions the controller never selects. Crucially, i and δ_t cannot depend on d .

Theorem 7.3 (Exact endogenous compression). *For a prescribed finite width profile S , the following are equivalent: a deterministic common design attains $R_d = v_d$ for every task; a privately randomized common design does so; a common design with an independent visible or hidden seed does so; and an endogenous compatible cover of profile S exists.*

Let $\beta_d > 0$ sum to one, and put $\mathcal{G} = \sum_d \beta_d (R_d - v_d)$. If the profile is infeasible, then every such design satisfies

$$\mathcal{G} \geq g_S := \beta_* \mu_* p_*^T \gamma_* > 0, \quad (7.4)$$

where $\beta_* = \min_d \beta_d$, μ_* is the least positive initial mass, p_* is the least positive transition probability, and γ_* is the least positive Bellman regret among supported entries. If there are no positive regrets, every profile with positive widths is feasible. The convention $p_* = 1$ applies when there are no transitions.

More generally, replacing the zero in (7.3) by the requirement $G_{t,d}(h, a_{t,d}(m)) \leq \eta_{t,d}$ gives a common design with $R_d - v_d \leq \sum_t \eta_{t,d}$. The sharper value is exactly the occupation-weighted sum in (7.2).

Proof. A cover defines the deterministic encoder $M_0 = i(H_0)$ and $M_{t+1} = \delta_t(M_t, A_t, Y_{t+1})$, and the task controller uses $a_{t,d}(M_t)$. Induction in (7.3) shows that every reached history lies in its indicated cell. Each incurred Bellman regret is zero, so Lemma 7.1 proves exactness. The same induction and identity prove the approximate-cover statement.

Conversely, an exact deterministic design defines $C_{t,m}^d$ to be the set of histories reached with positive probability under task d and state m . Initial coverage and common successor closure follow from its encoder. Since the nonnegative sum in (7.2) has zero expectation, every positively reached history has zero regret for its selected action. This proves all cover requirements. It also explains why adding closure for every unused action would impose a strictly different problem.

For a randomized design, Lemma 7.1 expresses the vector of excesses as an average of deterministic excess vectors. Each coordinate is nonnegative. If all averaged coordinates are zero, then, for almost every sampled design, all coordinates are zero, since the task set is finite. Choose one such deterministic table. This applies after conditioning on an arbitrary independent seed, with recovery from the seed and retained state when the seed is visible. It does not use a seed correlated with the initial history. The deterministic cover already constructed then proves necessity.

If no exact deterministic design exists, each deterministic design has some task, stage, and positively reached history with positive selected regret. That history has probability at least $\mu_* p_*^t$, and its regret is at least γ_* . Since $0 < p_* \leq 1$ and $t \leq T$, its contribution to $\mathcal{G}$ is at least the right side of (7.4). Every randomized design is an average of these vectors, so the same bound follows. When there is no positive regret, every action is Bellman optimal wherever supported, and a one-state encoder with arbitrary task actions is exact. $\square$

Example 7.4 (Why unused controls cannot define the cover). At the first time there is no report. Action **rest** leads to a deterministic hidden bit zero and a constant report, at zero cost. Action **probe** costs one, draws a fair bit, and reports it. After this report is consumed there is one blank checkpoint, followed by a binary guess with zero–one loss. The full-history optimum chooses **rest** and guesses zero; one retained state is exact. A restart family that insists on optimal guessing after the unused **probe** branch needs two retained states. The endogenous criterion and the all-controls restart criterion are both meaningful, but they have different feasible sets. The cover in Definition 7.2 implements the former, not the latter.

7.2. Lossy common design and intrinsic comparison. Let $\mathcal{D}_S^{\text{ctl}}$ be the finite set of deterministic common encoders and deterministic task controllers in this section. Put $r_d(\pi) = R_d(\pi) - v_d$. Then the hidden-selector lossy minimax value is

$$\min_{\lambda \in \Delta(\mathcal{D}_S^{\text{ctl}})} \max_d \sum_{\pi} \lambda_{\pi} r_d(\pi) = \max_{\beta \in \Delta(D)} \min_{\pi \in \mathcal{D}_S^{\text{ctl}}} \sum_d \beta_d r_d(\pi). \quad (7.5)$$

Here a selector chooses the entire design, including its task-controller tables; its mode is internal to that design. This follows from the simultaneous mixture identity in Lemma 7.1 and finite minimax. A mixture using at most $|D| + 1$ designs suffices and can be stored with at most $(|D| + 1)S_t$ private

labels. Private finite-width optimization is instead the polynomial optimization in the actual encoder and controller rows. The exact zero sets agree by Theorem 7.3, but positive minimax values need not agree. Thus the cover theorem supplies a distinguished zero face inside a quantitative lossy theory, rather than purporting to solve every lossy design problem by equivalence classes.

Corollary 7.5 (Intrinsic certification of endogenous widths). *Suppose the losses of this finite problem are normalized so that every total task loss lies in $[0, 1]$. Let an approximation F have the same control and task interfaces, with each task represented by the same bounded marked-transcript loss on both sides and the full-history benchmark included in the paired decision class. If*

$$\max\{\delta_1^P(E, F), \delta_1^P(F, E)\} \leq \varepsilon < g_S/4,$$

then the optimized β -weighted excess at profile S in F is less than $g_S/2$ exactly when the endogenous cover exists in E . If only a cost- K simulation from E to F is known and an exact S -state F design transports with zero error and the same benchmark, then an endogenous cover of profile KS exists in E . A hidden selector multiplies that profile by its charged selector size.

Proof. Full-history controllers also transport through a stateless simulator, so their optimized benchmarks differ by at most ε . Equation (2.18) bounds the optimized constrained values by ε in each direction. The optimized excesses therefore differ by at most 2ε . In E the value is zero on the cover face and at least g_S otherwise, by Theorem 7.3. The strict threshold follows. For the one-sided statement, the product construction transports the exact design at the stated width and benchmark. The same theorem then converts exactness into a cover. $\square$

Remark 7.6 (Dependence of the finite witness). The number g_S depends on the least positive probability and regret of the specified finite model. It can tend to zero as a transition or a cost is perturbed. Neither (7.4) nor the certification corollary asserts a model-uniform robustness radius. Likewise an inverse-limit assertion that every finite prefix has a feasible cover would give existence of an infinite consistent design, not a computable bound on the first infeasible prefix. No such bound is used here.

7.3. Task resolution, positive error, and the complete state. We give a quantitative state law at a source-consuming restart. It complements the endogenous zero-face theorem; it does not replace the causal update constraint by a terminal partition.

Let $\mathcal{H}$ be a finite set of supported histories at a fixed checkpoint. Choose d actual marked continuation-event prediction tasks, each including its declared open-loop continuation control word. For $h \in \mathcal{H}$ let $v(h) \in [0, 1]^d$ be the vector of their conditional probabilities. A source-consuming statistic has rows $Q(m | h)$, $m \in [K]$. For every preparation λ on $\mathcal{H}$, the task is announced after storage and its forecast is scored by squared error against the actual event. Define

$$e(Q) = \sup_{\lambda} \max_{1 \leq j \leq d} \mathbb{E}_{\lambda} \text{Var}(v_j(H) | M), \quad e_K = \inf_Q e(Q).$$

The expression is precisely excess Brier risk over the full-history forecast: conditional on H , event variance is an unavoidable common term, and the law of total variance gives the displayed residual. Decoders may use the announced preparation and task, but the same statistic Q must work for all of them. Let $N(\eta)$ be the maximum size of a subset of $v(\mathcal{H})$ whose Euclidean pair distances are at least $\eta > 0$. Let $C(r)$ be the minimum number of sets covering $v(\mathcal{H})$ with coordinatewise diameter at most r .

Theorem 7.7 (Positive-error state bounds). *For all positive K and η ,*

$$e_K \geq \frac{\eta^2}{2d} \left(1 - \frac{K}{N(\eta)}\right)_+. \quad (7.6)$$

For $0 \leq \varepsilon \leq 1/4$ there is a deterministic statistic of width $C(2\sqrt{\varepsilon})$ with $e(Q) \leq \varepsilon$. Consequently, writing $K(\varepsilon) = \min\{K : e_K \leq \varepsilon\}$,

$$N(\eta) \left(1 - \frac{2d\varepsilon}{\eta^2}\right) \leq K(\varepsilon) \leq C(2\sqrt{\varepsilon}). \quad (7.7)$$

At zero error the exact width is $|v(\mathcal{H})|$. It is at most the universal marked continuation width, and it equals that width when the event family separates all distinct marked continuation laws. The lower bound also holds when an independent exposed selector is available and each selector fiber has at most K private labels; in that case the conditional variance is taken given (U, M) .

Proof. Choose $n = N(\eta)$ separated histories and prepare them uniformly. For an output m , write $\pi_h(m) = \mathbb{P}(H = h \mid M = m)$. The variance identity gives

$$\begin{aligned} \sum_{j=1}^d \mathbb{E} \text{Var}(v_j(H) \mid M) &= \frac{1}{2} \sum_m \mathbb{P}(M = m) \sum_{h, h'} \pi_h(m) \pi_{h'}(m) \|v(h) - v(h')\|_2^2 \\ &\geq \frac{\eta^2}{2} \left(1 - \sum_m \mathbb{P}(M = m) \sum_h \pi_h(m)^2\right). \end{aligned}$$

Furthermore

$$\sum_m \mathbb{P}(M = m) \sum_h \pi_h(m)^2 \leq \sum_m \mathbb{P}(M = m) \max_h \pi_h(m) = \frac{1}{n} \sum_m \max_h Q(m \mid h) \leq \frac{K}{n}.$$

At least one coordinate variance is the average of these d terms or larger. Nonnegativity proves (7.6). Conditioning on an independent exposed selector gives the identical bound on the sum of coordinate variances at every fiber, with the same uniform preparation. Average this sum over the selector and only then choose a coordinate at least as large as its average. Thus the task need not depend on the selector, and the same lower bound follows.

For the upper bound assign each history to a covering set containing its feature vector and retain its set index. Every conditional coordinate range then has length at most $2\sqrt{\varepsilon}$. A real random variable in $[a, b]$ has variance at most $(b - a)^2/4$: $\mathbb{E}(X - a)(b - X) \geq 0$ implies $\text{Var } X \leq (\mathbb{E}X - a)(b - \mathbb{E}X) \leq (b - a)^2/4$. This proves the upper bound uniformly over preparations.

If two histories with different feature vectors share a retained symbol with positive probability, a preparation on just those two histories has a strictly positive posterior variance at that symbol in a differing coordinate. Therefore zero excess forces disjoint retained supports for distinct feature vectors. Conversely the feature vector itself is an exact statistic. This proves the zero law. Every feature is determined by the universal marked continuation law; a finite separating event family distinguishes all its finitely many classes. The final assertion follows. $\square$

Corollary 7.8 (A positive-error obstruction for marked simulation). *Suppose a prescribed source-consuming reverse simulator of width K compares the stored-state restart to the full-history restart, preserving the actual marked continuation experiment with error at most δ uniformly over preparations and announced continuation tasks. Then*

$$\delta \geq e_K \geq \frac{\eta^2}{2d} \left(1 - \frac{K}{N(\eta)}\right)_+.$$

Proof. Compose the reverse simulator with the full-history optimal Brier forecast for each announced task. Squared error belongs to $[0, 1]$; marked comparison changes its expectation by at most δ . The optimal stored-state forecast is no worse than this composed forecast. Hence its excess is at most δ for every preparation and task. Apply Theorem 7.7. $\square$

For several checkpoints these are necessary width bounds at every restart. Their simultaneous sufficiency additionally requires that the chosen covers admit causal updates on the reached

support, exactly as in the preceding endogenous theorem. A feature cover alone does not construct a reverse channel preserving an arbitrary actual mark. For a full separating family the zero-error envelope recovers the universal continuation quotient; before completion, the quantitative bounds depend on the declared tasks rather than on irrelevant marks or unused future controls.

8. REGENERATIVE AVERAGE RISKS

8.1. The reset signature. Let the physical state space X be standard Borel, with any actual persistent hidden mark included in X . Reports and actions are finite. A stationary common finite-register design consists of stochastic encoder rows and task-controller rows, independent of time; their product of simplexes is a compact set Π_S . A fixed finite task set is part of the row index $j \in J$, where J is compact metrizable. For each design and row, the joint physical-register transition has form

$$P_j^\pi = \eta N_j + (1 - \eta) Q_j^\pi, \quad 0 < \eta \leq 1, \quad N_j(x, m; \cdot) = \nu_j \otimes \delta_{m_0}. \quad (8.1)$$

The independent reset flag is physically supplied; it refreshes the physical preparation and sets the within-policy memory to m_0 . Thus N_j is a constant kernel independent of π . Actions between resets may change the physical state. No domination of Q_j^π by one reference measure is assumed.

Let $c_j^\pi : X \times [S] \rightarrow [0, 1]$ be the expected one-stage marked cost. We assume the following uniform continuity conditions, using the supremum over starting states for kernel total variation:

$$(j, \pi) \mapsto Q_j^\pi \text{ is continuous in } \sup_{x, m} \|\cdot\|_{\text{TV}}, \quad j \mapsto \nu_j \text{ in TV}, \quad (j, \pi) \mapsto c_j^\pi \text{ in } \|\cdot\|_\infty. \quad (8.2)$$

For fixed physical kernels with finite action and report alphabets, continuity in π follows directly by expanding the finite sums of encoder and controller probabilities; the physical kernels can be singular. Row dependence must independently satisfy (8.2). In particular changing a deterministic flow with the row does not automatically satisfy a total-variation continuity hypothesis.

Theorem 8.1 (Nonsummable average-risk duality). *Under (8.1)–(8.2), every (j, π) has a unique invariant probability ζ_j^π . Its average risk*

$$r_\pi(j) = \zeta_j^\pi c_j^\pi$$

is the limit of expected Cesàro average cost from every initial law. For $N \geq 1$ and $0 < \beta < 1$ the following errors are uniform in j, π , and the initial law ξ :

$$\left| \frac{1}{N} \sum_{t=0}^{N-1} \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1}{N\eta}, \quad (8.3)$$

$$\left| (1 - \beta) \sum_{t=0}^{\infty} \beta^t \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1 - \beta}{1 - \beta(1 - \eta)}. \quad (8.4)$$

The map $\pi \mapsto r_\pi$ is continuous into $C(J)$ and has compact image. The attainable average-risk body under independent selection of a stationary design is compact and convex in $C(J)$. For continuous $b : J \rightarrow \mathbb{R}$ its common-design minimax value satisfies

$$\min_{\Lambda \in \mathcal{P}(\Pi_S)} \max_{j \in J} \left(\int r_\pi(j) d\Lambda(\pi) - b(j) \right) = \max_{w \in \mathcal{P}(J)} \min_{\pi \in \Pi_S} \int_J (r_\pi(j) - b(j)) dw(j), \quad (8.5)$$

and all displayed extrema are attained. For finite J a selector of at most $|J| + 1$ values realizes every required optimum. More generally Proposition 2.5 gives an intrinsic finite-dimensional selector approximation.

Proof. A probability kernel contracts total variation. Subtracting two probabilities in (8.1) cancels the common reset part, giving

$$\|\xi P_j^\pi - \xi' P_j^\pi\|_{\text{TV}} \leq (1 - \eta) \|\xi - \xi'\|_{\text{TV}}.$$

The explicit probability

$$\zeta_j^\pi = \eta \sum_{k=0}^{\infty} (1-\eta)^k (\nu_j \otimes \delta_{m_0}) (Q_j^\pi)^k \quad (8.6)$$

is invariant by direct substitution, with convergence in total variation. Contraction proves uniqueness and $\|\xi(P_j^\pi)^t - \zeta_j^\pi\|_{\text{TV}} \leq (1-\eta)^t$. Integrating the bounded cost and summing the geometric series proves (8.3) and (8.4), including their uniformity.

For two parameter-policy pairs let P, P' denote their kernels and ζ, ζ' their invariant laws. Invariance, the preceding contraction, and a one-step difference give

$$\|\zeta - \zeta'\|_{\text{TV}} \leq (1-\eta) \|\zeta - \zeta'\|_{\text{TV}} + \sup_{x,m} \|P(x, m; \cdot) - P'(x, m; \cdot)\|_{\text{TV}}.$$

Thus the invariant-law difference is at most the last supremum divided by η . Together with the cost difference, this proves joint continuity of $(j, \pi) \mapsto r_\pi(j)$. Compactness of $J \times \Pi_S$ upgrades it to continuity of $\pi \mapsto r_\pi$ in the uniform norm. Its image is compact.

Probabilities on compact Π_S form a weakly compact set. The barycenter map into $C(J)$ is norm continuous: approximate the compact risk image uniformly by finitely many continuous coordinates, or use uniform equicontinuity of $r_\pi(j)$ and a finite net in J . The image is therefore compact and convex. The payoff $\int (r_\pi - b) d\Lambda dw$ is continuous and affine in both probabilities on compact spaces. Convex minimax gives (8.5). The inner minimum over a probability on Π_S equals the minimum over π , and compactness attains it. A finite-dimensional barycenter is reduced by the affine-dependence argument already proved in Theorem 2.2. This proves the finite selector assertion. $\square$

Remark 8.2 (Which policies are optimized). The theorem optimizes stationary finite-register common policies with the reset signature (8.1). It does not assert that a stationary policy is optimal among arbitrary nonstationary policies without that signature. A selector choosing a stationary policy is sampled once; its mode persists through resets while that mode's within-policy register is reset. If the selector must be private, store the mode and the register, paying their product. Conditioned on the mode, (8.3)–(8.4) hold with the same constants, so they also hold after averaging. The joint chain including a persistent mode need not have a unique invariant law across modes, and no such claim is needed.

8.2. Cycle comparison before optimization. Consider two reset experiments with the same reset probability and reset flag. A comparison within a fresh cycle starts with the specified physical preparation and initial register. Suppose, for every feedback design and every independent visible side seed, the joint laws of the marked information needed to evaluate the age- k stage cost differ by at most $\varepsilon_k \leq 1$. Here age zero means the first stage following a reset; its law includes any reports used at that stage. The same control law and encoder rows are used on both sides. This is a uniform channel comparison, not a comparison obtained after separately optimizing policies.

Theorem 8.3 (Regenerative resource transport). *The resulting paired average risks satisfy, for every consumer width,*

$$\|r_E - r_F\|_\infty \leq \mathfrak{e} := \eta \sum_{k=0}^{\infty} (1-\eta)^k \varepsilon_k. \quad (8.7)$$

The bound holds with a priced simulator by replacing the consumer width S by its product with the simulator width and by charging the selector as in Theorem 2.2. For a sequence of stateless two-sided comparisons with $\varepsilon_{n,k} \rightarrow 0$ for each fixed k , $\mathfrak{e}_n \rightarrow 0$, uniformly over all consumer widths. At every width, the corresponding average-risk bodies and optimized lower support values have the same error bound. The minimax values in (8.5) with the same centering b differ by at most $\mathfrak{e}_n$.

Proof. Formula (8.6) expresses a stationary average risk as the geometric mixture of costs at each cycle age. The expectation of a $[0, 1]$ marked stage cost changes by at most ε_k in that age's joint law, yielding (8.7). The proof is uniform in the consumer because its error hypothesis is uniform before integration. For a priced transducer, include its retained state in the joint register and reset it to its specified initial state at the cycle boundary. The same product construction and conditioning on independent mode choices applies.

The geometric weights sum to one. Given $\delta > 0$, first choose L so their tail beyond L is less than $\delta/2$, and then choose n so the finite weighted sum through L is less than $\delta/2$. This proves $\epsilon_n \rightarrow 0$ without a rate uniform in k . Pairing each design with the same design on the other side gives the risk-body comparison. Taking weighted infima or the supremum of centered row risks proves the optimized statements; taking mixtures preserves the same bound. $\square$

Example 8.4 (A nondominated physical kernel). On $X = [0, 1]$, let $Q^a(x, \cdot) = \delta_{(x+a) \bmod 1}$ for finitely many rotations a , and let ν be Lebesgue probability. The transition $\eta\nu + (1 - \eta)Q^a(x, \cdot)$, as x ranges over X , is not dominated by a common σ -finite measure when $\eta < 1$: such a measure would have to give positive mass to uncountably many singleton points. A σ -finite measure has at most countably many positive-mass atoms, by decomposing into finite-measure sets and positive lower bounds on atomic mass. Finite report partitions and any common finite encoder fit Theorem 8.1. This is a nonsummable controlled risk theorem with a genuinely nondominated physical transition family, not an application of full-path density compactness.

Remark 8.5 (Why the reset is explicit). A hidden physical process may mix while a retained controller bit never changes. Physical mixing alone then gives no uniform contraction on the joint physical–register space. Equation (8.1) requires an actual clearing of that register. Conversely, without a cycle tail or another uniform long-time condition, finite-prefix comparison errors need not control average cost: a sequence of processes that agrees with a zero-cost process for its first n stages and incurs unit cost forever afterward has vanishing error at each fixed prefix but average cost one. The positive theorem uses the explicit geometric cycle law to control this precise obstruction.

9. AN ACTIVE MICROSCOPIC REALIZATION

9.1. Interventions and the unbounded flow. Fix a nonempty finite system of $N \geq 2$ identical hard spheres of positive diameter on a flat torus, with positive configuration volume. Let Γ_N be the admissible phase space and let μ be the normalized measure with density proportional to $\mathbf{1}_{\Gamma_N}(q, v) \exp(-b \sum_i |v_i|^2)$, $b > 0$. We use the classical almost-everywhere hard-sphere flow Φ_t , which is an invertible measure-preserving flow off a null exceptional set; see [6] and the detailed microscopic construction retained in the companion. No claim about particle-number uniformity is needed. On $H = L^2(\mu)$,

$$U(t)f = f \circ \Phi_t$$

is a strongly continuous unitary group, with skew-adjoint generator L . Indeed invariance gives isometry and invertibility; for bounded continuous phase functions, almost-everywhere continuity of the orbit at zero and dominated convergence give strong continuity. Density and isometry extend it to H . The generator domain includes the collision identifications of the flow; it is not the unrestricted formal transport derivative on arbitrary boundary data.

Let A be finite and choose orthogonal matrices O_a on velocity space. The intervention

$$R_a(q_1, \dots, q_N, v_1, \dots, v_N) = (q_1, \dots, q_N, O_a v_1, \dots, O_a v_N)$$

is an invertible μ -preserving map. Put $C_a f = f \circ R_a$. For a fixed step length $\Delta > 0$, the controlled physical step is $F_a = \Phi_\Delta \circ R_a$, and its Koopman operator is

$$T_a = C_a U(\Delta). \tag{9.1}$$

The order matters. If both $O_a = I$ and $O_a = -I$ are allowed, the two steps give different subsequent positions on a positive-measure set of configurations with no collision in the next Δ units of time and nonzero sufficiently small velocities. Such a set exists inside any nonempty interior admissible configuration region by choosing small velocities and then a positive interval away from zero. The action therefore changes the dynamics, rather than merely choosing which observable to read. We require no assertion that $C_a D(L)$ is contained in $D(L)$.

For all finite control words, the flow and intervention compositions are well defined on a common full-measure subset: the preimage of a null exceptional flow set under each finite measure-preserving composition is null, and there are countably many finite words. Every preparation absolutely continuous with respect to μ is supported there, including under a feedback choice of the word.

9.2. A graph-core approximation and controlled words. Choose an increasing family of finite-dimensional real subspaces $V_n \subset D(L)$ whose union is a graph core, and let P_n be the orthogonal projections. Such a family can be constructed without postulating smooth collision-boundary coordinates. Start with a countable dense family (f_j) in H and a smooth compactly supported approximate identity χ_ε on $\mathbb{R}$. Define

$$f_{j,\varepsilon} = \int_{\mathbb{R}} \chi_\varepsilon(s) U(s) f_j ds.$$

A difference quotient followed by translation of χ_ε shows that $f_{j,\varepsilon} \in D(L^k)$ for every k and

$$L^k f_{j,\varepsilon} = (-1)^k \int \chi_\varepsilon^{(k)}(s) U(s) f_j ds, \quad \|L^k f_{j,\varepsilon}\|_2 \leq \|\chi_\varepsilon^{(k)}\|_1 \|f_j\|_2. \quad (9.2)$$

For $f \in D(L)$ its smoothed versions converge to f in graph norm: commuting L on its domain reduces this to strong continuity for f and Lf . At each fixed smoothing scale, approximation of f by the dense family controls both the function and its first derivative by (9.2). A countable sequence of scales and finite linear spans consequently gives a graph core. Enumerate those vectors and let V_n be their successive spans. This also shows that their collision compatibility comes from the actual flow domain, not from an imposed incompatible trace.

Set $L_n = P_n L P_n$ on V_n , $U_n(t) = \exp(tL_n)$ on V_n , and, as an operator on H extended by zero on $V_n^\perp$, define

$$T_{a,n} = P_n C_a P_n U_n(\Delta) P_n. \quad (9.3)$$

These operators are contractions. Their finite-dimensional matrices need not be Markov matrices: they approximate *observables*; the observation experiment constructed below is nevertheless a genuine probability experiment.

Lemma 9.1 (Graph-core and controlled-word convergence). *For every $f \in H$ and $T_0 < \infty$,*

$$\sup_{0 \leq t \leq T_0} \|U_n(t) P_n f - U(t) f\|_2 \longrightarrow 0. \quad (9.4)$$

For every fixed word length k ,

$$\max_{(a_1, \dots, a_k) \in A^k} \|T_{a_1, n} \cdots T_{a_k, n} f - T_{a_1} \cdots T_{a_k} f\|_2 \longrightarrow 0. \quad (9.5)$$

Neither statement needs domain invariance under the interventions.

Proof. The matrix L_n is skew-adjoint on V_n because L is skew-adjoint and $V_n \subset D(L)$. Thus its group is unitary and $\|(z - L_n)^{-1}\| \leq 1/\Re z$ when $\Re z > 0$. If $u \in V_m$ and $n \geq m$, then

$$(z - L_n)^{-1} P_n (z - L) u = u.$$

The set $(z - L) \bigcup_m V_m$ is dense in H , since the union is a graph core and $z - L$ maps $D(L)$ onto H . The uniform resolvent bounds and $P_n \rightarrow I$ strongly therefore give

$$(z - L_n)^{-1} P_n f \longrightarrow (z - L)^{-1} f \quad (f \in H).$$

This is the graph-core consistency and stable-resolvent setting of Trotter's theorem [9, Theorem 5.2]. We give the contraction argument to specify the hypotheses used here. For any contraction generator A and $u \in D(A)$, the Euler resolvent formula can be written as a Gamma integral:

$$(I - tA/m)^{-m}u = \int_0^\infty e^{sA}u d\gamma_{m,t}(s), \quad \mathbb{E}_{\gamma_{m,t}}s = t, \quad \text{Var}_{\gamma_{m,t}}s = t^2/m.$$

This follows by multiplying the Laplace resolvent integrals and convolving their exponential densities. Since $\|e^{sA}u - e^{tA}u\| \leq |s - t|\|Au\|$, its error from $e^{tA}u$ is at most $t\|Au\|/\sqrt{m}$. For $u \in V_l$ one has $\|L_n u\| \leq \|Lu\|$ for all $n \geq l$. At each fixed $t > 0$ and Euler order m , the resolvent convergence already proved gives convergence of its m th power. First letting $n \rightarrow \infty$ and then $m \rightarrow \infty$ proves orbit convergence on the core. The orbit Lipschitz bound with constant $\|Lu\|$, uniform in n , upgrades it to uniformity on $[0, T_0]$ by a finite time mesh. Density and contraction extend it to all $f \in H$. No derivative of $U(t)f$ for $f \notin D(L)$ was taken.

Now $P_n C_a P_n \rightarrow C_a$ strongly by boundedness of C_a and strong convergence of P_n . Equation (9.4) gives $T_{a,n} \rightarrow T_a$ strongly. For a fixed finite word, subtract the two products one factor at a time. Each difference acts on a fixed vector formed from the limiting factors, while every factor to its left is a contraction. Each term tends to zero. There are finitely many words of length k , which proves the maximum in (9.5). At no step was an expression such as $LC_a f$ required. $\square$

9.3. Actual marked acquisition under feedback. Fix real observables $f_k \in H$ and noises $\sigma_k > 0$. At step k , a controller chooses a_k from its previously available reports and retained state, the action is applied to the physical state, and the new report is

$$Y_k = (T_{a_1} \cdots T_{a_k} f_k)(x_0) + \sigma_k \xi_k, \quad \xi_k \text{ independent standard Gaussians.} \quad (9.6)$$

The initial law is $p_\theta d\mu$, with $\sup_\theta \|p_\theta\|_2 \leq D < \infty$. An unobserved initial physical mark W is drawn jointly with x_0 through a fixed channel $\gamma(dw | x_0)$, independently of the acquisition noises. Losses may use that actual mark and the observed control-report transcript. This specification does not allow an unproved replacement of a controlled terminal physical quantity by an independent variable of the same marginal law.

The approximation replaces the product in (9.6) by $T_{a_1,n} \cdots T_{a_k,n} f_k$. It has a finite-dimensional physical realization. In a real orthonormal basis $(\varphi_1, \dots, \varphi_{d_n})$ of V_n , write the matrix of $T_{a,n}$ as $M_{a,n}$ and the coefficient vector of $P_n f_k$ as $c_{k,n}$. The initial row vector is $v_0 = (\varphi_1(x_0), \dots, \varphi_{d_n}(x_0))$, and

$$v_k = v_{k-1} M_{a_k,n}, \quad Y_k^{(n)} = v_k c_{k,n} + \sigma_k \xi_k.$$

Initialize (v_0, W) with the actual push-forward joint law of (x_0, W) . This is a valid controlled probability model with a real vector state; no positivity of $M_{a,n}$ is needed. Its real dimension d_n is not counted as d_n finite retained labels. The consumer register remains a separate constrained object. Measurable representatives of the finitely many functions may be chosen once; their null ambiguities have zero probability under the specified preparations.

Theorem 9.2 (Active marked deficiency bound). *For each fixed horizon T , define*

$$\epsilon_{n,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \sum_{a \in A^k} \|T_{a_1} \cdots T_{a_k} f_k - T_{a_1,n} \cdots T_{a_k,n} f_k\|_2. \quad (9.7)$$

The exact and approximate acquisitions have two-sided stateless marked causal deficiency at most $\epsilon_{n,T}$, using the identity report channel. The bound includes every independent visible seed in the joint comparison, holds uniformly over every finite-register consumer and feedback design, and tends to zero for each fixed T . Consequently all paired $[0, 1]$ task risks and optimized lower support values have error at most $\epsilon_{n,T}$ at unchanged widths; paired centered values have error at most $2\epsilon_{n,T}$ when their unrestricted benchmarks are also paired.

Proof. Couple the two experiments with the same x_0, W and every independent visible seed. Couple their private controller randomness so that the same available history produces the same action. At each step before the first report mismatch, use a maximal coupling of the two Gaussian reports. Gaussian densities satisfy

$$\|N(u, \sigma^2) - N(v, \sigma^2)\|_{\text{TV}} = 2\Phi(|u - v|/(2\sigma)) - 1 \leq \frac{|u - v|}{\sigma\sqrt{2\pi}}.$$

The equality follows by separating the densities at their midpoint; the inequality follows by bounding the standard normal density by $1/\sqrt{2\pi}$. Maximal couplings can be chosen measurably from these densities by their common part and residual parts.

For a fixed x_0 , the probability of a first mismatch at time k is bounded by the sum, over the possible matched action words $a \in A^k$, of their matched-history probabilities times the displayed Gaussian bound for that word. Discarding these probabilities gives an upper bound by the sum over all words. Integrate under the *original* preparation $p_\theta d\mu$. Cauchy–Schwarz bounds the integral of each absolute mean difference by D times its L^2 norm. Summing first-mismatch probabilities yields (9.7). In particular no bound on a feedback-conditioned posterior density has been assumed. When no mismatch occurs the entire transcript and actual mark, including the visible seed, coincide. The coupling inequality gives their joint total variation bound.

The same argument is symmetric in the two report models; the stateless identity channel is executable in both directions, so it bounds the intrinsic infima. Lemma 9.1 makes every term in the fixed finite sum tend to zero. The comparison precedes any consumer optimization, hence its bound does not depend on the consumer’s number of labels. The remaining conclusions are the channel and support inequalities in Theorem 2.2 and (2.18), which remain valid on these standard Borel carriers. $\square$

9.4. Active regenerative average decisions. Take a fixed observable f , fixed noise $\sigma > 0$, and a fixed finite measurable report partition applied to each Gaussian acquisition. At each cycle boundary the physical system is re-prepared from $p_\theta d\mu$ with its actual mark channel γ , and the within-policy register is reset to m_0 . Between boundaries the orthogonal interventions and hard-sphere flow continue as above. The independent reset probability is $\eta > 0$. In the approximate system reset the real vector and mark by the same joint push-forward preparation. A stationary consumer uses the partitioned reports, actions, and its finite register. Its $[0, 1]$ one-stage losses depend on the actual cycle mark, control, and current finite report; reset stage costs are fixed and identically paired. A finite task index is available to the controller but not to the common encoder.

Assume $j \mapsto p_j$ is continuous in $L^1(\mu)$ on a compact row set, the L^2 bounds are uniform, and the bounded row losses vary uniformly continuously. The flow, intervention maps, observable, noise law, report partition, and mark channel are fixed across rows. Then the exact and every approximate model satisfy (8.2): only the reset law and losses vary with the row, and the dependence on finite policy rows is a finite polynomial expansion in kernels of norm one. The underlying physical transitions are allowed to be deterministic between resets.

Corollary 9.3 (Active average-risk realization). *Both models have the compact average-risk bodies and attained duality of Theorem 8.1. At every finite consumer width their paired average risks, lower supports, and equally centered minimax values differ by at most*

$$\epsilon_n = \eta \sum_{k=0}^{\infty} (1 - \eta)^k \epsilon_{n,k+1} \longrightarrow 0. \quad (9.8)$$

The convergence is uniform in the consumer width and in any independent priced mixture of stationary designs. It does not require a controlled word approximation rate uniform in word length.

Proof. A deterministic report partition contracts total variation. The age- k stage is evaluated after at most $k + 1$ controlled acquisitions, so Theorem 9.2 supplies the cycle comparison with $\varepsilon_{n,k} = \varepsilon_{n,k+1}$. Apply Theorem 8.3. Its geometric tail proof gives the limit although the finite-word bound in (9.7) can grow with k . The reset and row-continuity assertions just verified give Theorem 8.1. Conditioning on each persistent mode and charging its index proves the mixture statement. $\square$

Remark 9.4 (Fixed physical scope). The interventions above are active changes to microscopic dynamics, and the average theorem uses actual repeated physical preparation. Neither observation control alone nor an assumed posterior L^2 bound is substituted for them. The assertions are for fixed N , sphere diameter, A , Δ , and positive noise, and for the declared initial-mark losses. No estimate is asserted uniform in particle number, a growing action alphabet, vanishing noise, or a kinetic scaling. The historical nonlinear kinetic corrector and the Sinai left-strip spectral problem remain different analytic statements. They are preserved in the development with their original hypotheses, not inferred from the present operator approximation.

9.5. A posteriori bounds from graph and intervention residuals. Strong controlled-word convergence supplies an asymptotic comparison. For a fixed approximation space one can also bound its actual comparison error using finite Gram data. We retain the physical setup of Theorem 9.2: the particle number, interventions, noise, step length, and actual initial-mark channel are fixed. These statements do not require the interventions to map $D(L)$ into itself.

Suppress n and let $V \subset D(L)$ have orthonormal basis $\phi_1, \dots, \phi_d$. Identify V with $\mathbb{R}^d$ by an isometry ι ; complexification gives the same formulas with adjoints. Write

$$P = \iota^*, \quad J = \iota^* L \iota, \quad B_a = \iota^* C_a \iota, \quad M_a = B_a e^{\Delta J}.$$

Because L is skew-adjoint and C_a is unitary, J is skew-adjoint, e^{sJ} is unitary and M_a is a contraction. The two residual Gram matrices are

$$R = (L \iota)^* L \iota - J^* J = ((I - P) L \iota)^* ((I - P) L \iota), \quad (9.9)$$

$$S_a = I - B_a^* B_a = ((I - P) C_a \iota)^* ((I - P) C_a \iota). \quad (9.10)$$

In particular both are positive semidefinite. For a coefficient vector c , set

$$d_a(c) = \int_0^\Delta \sqrt{(e^{sJ} c)^* R (e^{sJ} c)} ds + \sqrt{(e^{\Delta J} c)^* S_a (e^{\Delta J} c)}. \quad (9.11)$$

All quantities in this formula are finite-dimensional. For a real observable $f \in H$ put

$$c_f = \iota^* f, \quad \tau_f = (\|f\|_H^2 - \|c_f\|^2)^{1/2}.$$

If $w = (a_1, \dots, a_k)$ and $c_j = M_{a_{j+1}} \cdots M_{a_k} c_f$ (empty product equal to the identity), define

$$E_w(f) = \tau_f + \sum_{j=1}^k d_{a_j}(c_j). \quad (9.12)$$

The suffix order agrees with the Koopman product for a physically chronological word; its reversal would approximate a different dynamics.

Theorem 9.5 (Residual control of the active marked experiment). *For every such space V and every controlled word,*

$$\|T_{a_1} \cdots T_{a_k} f - \iota M_{a_1} \cdots M_{a_k} c_f\|_H \leq E_w(f). \quad (9.13)$$

For the preparations p_θ with $\sup_\theta \|p_\theta\|_2 \leq D$ and Gaussian acquisitions of Theorem 9.2, the two stateless marked feedback deficiencies are bounded by

$$\hat{\epsilon}_{V,T} = \min \left\{ 1, \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left(\sum_{w \in A^k} E_w(f_k)^2 \right)^{1/2} \right\}. \quad (9.14)$$

The comparison includes every independent exposed seed and is uniform over all finite consumer widths. Thus it bounds raw paired risks and lower supports at unchanged widths. When unrestricted benchmarks are also paired, the centered risk bound is $2\hat{\epsilon}_{V,T}$.

Proof. The orthogonal splitting of $L\iota c$ into its V component ιJc and its orthogonal component proves (9.9). Unitarity of C_a similarly proves (9.10). For $v = \iota c$, the path $s \mapsto \iota e^{sJ}c$ lies in the finite space $V \subset D(L)$ and is differentiable in the graph norm. Hence variation of constants is legitimate and gives

$$U(\Delta)v - \iota e^{\Delta J}c = \int_0^\Delta U(\Delta - s)(L\iota - \iota J)e^{sJ}c ds.$$

Taking norms and applying (9.9) bounds this difference by the integral in (9.11). Apply the unitary C_a and then add the orthogonal projection error $\|(I - P)C_a \iota e^{\Delta J}c\|$ to obtain

$$\|T_a \iota c - \iota M_a c\| \leq d_a(c).$$

No derivative of $C_a \iota c$ has occurred. First remove $(I - P)f$ at cost τ_f . Telescope the difference of the two word products with exact prefixes and approximate suffixes. All prefixes are isometries and all approximate suffixes are contractions. The j th term is bounded by $d_{a_j}(c_j)$, proving (9.13).

For the probabilistic assertion use the common initial x_0 , its actual conditional mark W , and the same visible seed, as in Theorem 9.2. Before the first report mismatch the selected actions agree. Conditional on x_0 , the weights of the matched histories leading to the possible words of length k sum to at most one. If $e_w(x)$ is the corresponding exact-minus-approximate mean, the first-mismatch contribution at time k is therefore at most

$$\frac{1}{\sigma_k \sqrt{2\pi}} \int p_\theta(x) \max_{w \in A^k} |e_w(x)| d\mu(x).$$

This retains the history weights until after taking their maximum, instead of replacing each by one. Cauchy–Schwarz and $\max_w |e_w|^2 \leq \sum_w |e_w|^2$ bound the integral by $D(\sum_w \|e_w\|_2^2)^{1/2}$. Apply (9.13) and sum the first-mismatch probabilities. The coupling is symmetric and preserves the whole actual marked transcript on its matching event, so it proves both stateless inequalities in (9.14). It conditions only on the original preparation, not on a feedback posterior with a supposed L^2 bound. Conditioning on an independent exposed seed gives the same uniform estimate. Risk and support statements follow by composition with any consumer and by taking the appropriate infima. $\square$

Corollary 9.6 (Finite-data average-risk and presentation certificates). *In the active regenerative model of Corollary 9.3, a finite age cutoff H gives an error bound for paired average risks and lower supports, uniform in consumer width,*

$$\hat{\epsilon}_{V,H} = \eta \sum_{k=0}^H (1 - \eta)^k \hat{\epsilon}_{V,k+1} + (1 - \eta)^{H+1}. \quad (9.15)$$

The same bound applies to identically centered values; separately paired unrestricted benchmarks give twice the bound. Finite-horizon bounds (9.14), combined with any other verified stateless presentation bridges, can be inserted into (5.13) without changing the intrinsic resource budget.

Proof. At age k there are at most $k+1$ acquisitions. Use Theorem 9.5 at those horizons and bound each remaining age error by one. The remaining geometric mass is exactly $(1-\eta)^{H+1}$. The cycle formula in Theorem 8.3 proves (9.15). The final assertion is Theorem 5.6. A finite-dimensional vector model is not automatically a finite report experiment; a further finite-report bridge must actually be proved before invoking the rational local testing algorithm. No such bridge is implicit in a matrix representation. $\square$

The data required in (9.9)–(9.12) are inner products of $f, \phi_i, L\phi_i, C_a\phi_i$. Once these data have rigorous enclosures, the formulas give a posteriori bounds without knowing the exact controlled orbit. For example, certified numbers $r_0 \geq \|R\|$ and $s_{a,0} \geq \|S_a\|$ imply the simpler bound

$$d_a(c) \leq (\Delta\sqrt{r_0} + \sqrt{s_{a,0}})\|c\|.$$

With matrix-entry enclosures, interval arithmetic and this inequality avoid an uncertified numerical integral. Floating-point Gram entries alone are not an analytic certificate. Nor is it asserted that these residual majorants converge for every graph-core sequence: cancellation may make the actual orbit error small while its residual bound is large. The convergence theorem remains Lemma 9.1; Theorem 9.5 is a complementary finite-space bound. This distinction separates a proved limiting approximation from a checkable error estimate for a particular space.

10. FINITE RATIONAL PRESENTATIONS OF GAUSSIAN ACQUISITIONS

We now join finite resource certification to the actual marked physical experiment. The intermediate model is allowed to have a nonfinite latent state. Its reports, not merely its observable coordinates, must eventually be finite. Throughout this section the parameter, action and mark alphabets are finite, the horizon is finite, and the mark is drawn with the initial state. An initial report, when present, is treated as one additional acquisition. All channels below are parameter blind and task blind.

10.1. A stateless report channel. Let (X, μ) be a probability space, let the preparations be $p_\theta\mu$ with $\|p_\theta\|_2 \leq D$, and let $\gamma(dw \mid x)$ be the actual initial-mark channel. For each action word $w \in A^k$, let $m_{k,w} \in L^2(\mu)$ be a real mean function. The report at time k is

$$Y_k = m_{k,a_{1:k}}(x) + \sigma_k \xi_k, \quad \sigma_k > 0,$$

where the noises are independent standard Gaussians, independent of (x, W) . Actions may be selected from all previously emitted reports. The finite-dimensional models in the preceding section have this form; so do the enriched approximations in Section 11. Put

$$B_k = \left(\sum_{w \in A^k} \|m_{k,w}\|_2^2 \right)^{1/2}.$$

Choose $R > 0$ and a finite interval partition of $[-R, R]$ with maximum cell length h . Adjoin the two tail cells. Let Q return the cell of a real report. Let H reconstruct an interior cell uniformly and reconstruct the tail cells at $-R$ and R , respectively. The endpoints may be assigned arbitrarily to adjacent cells. These maps use no persistent memory, seed, parameter or mark. Fresh randomness in H is transient. Denote by $F^{R,h}$ the physical acquisition with the report $Q(Y_k)$; its latent state and actual mark are unchanged.

Lemma 10.1 (Uniform marked report quantization). *The channel Q simulates $F^{R,h}$ from F exactly. The reverse channel H has marked feedback error at most*

$$\beta(R, h) = 1 \wedge \sum_{k=1}^T \left\{ \frac{h}{\sigma_k \sqrt{2\pi}} + \frac{2DB_k}{R} + \frac{4\sigma_k^2}{R^2} \right\}. \quad (10.1)$$

The comparison includes the joint law of every independent exposed selector and is uniform over every finite consumer width. In particular $h \downarrow 0$ and $R \uparrow \infty$ give two-sided stateless marked comparison with error tending to zero.

Proof. For an absolutely continuous density g and an interval I of length at most h , write $g_I = |I|^{-1} \int_I g$. The fundamental theorem of calculus and Fubini give

$$\int_I |g - g_I| \leq |I|^{-1} \int_{I^2} |g(u) - g(v)| du dv \leq h \int_I |g'|.$$

Consequently the variation error of the interior histogram is at most $\frac{h}{2} \int |g'|$. For a Gaussian of variance σ^2 the latter integral is $\sqrt{2/\pi}/\sigma$. Replacement of the tails costs at most their total probability, even though the reconstruction is atomic. For any mean m , this probability is at most

$$\mathbf{1}_{\{|m| > R/2\}} + \mathbb{P}\{|\sigma\xi| > R/2\} \leq \frac{2|m|}{R} + \frac{4\sigma^2}{R^2}.$$

Thus the conditional report error, given the original x and the selected action word, has the three terms in (10.1) before integration.

Couple the exact and reconstructed systems with the same original (x, W) and independent selector. Before their first emitted-report mismatch use the same controller randomness and maximally couple the next reports. Matched-history weights of different words at time k sum to at most one. Therefore the mean term is bounded by

$$\frac{2}{R} \int p_\theta(x) \max_{w \in A^k} |m_{k,w}(x)| d\mu(x) \leq \frac{2D}{R} \left(\sum_w \|m_{k,w}\|_2^2 \right)^{1/2}.$$

This uses the original preparation, not an assumed bound for a feedback-conditioned posterior. Sum the first-mismatch bounds. On the matching event the complete emitted transcript, controls, mark and selector agree. This proves the reverse comparison. Applying Q to each original report is the exact forward simulation, also under feedback. Conditioning on the exposed selector throughout proves the joint assertion. Composition never enlarges a retained register. $\square$

The bound is deliberately an elementary certified bound; Gaussian tail integrals or higher moments may improve it. No uniform bound on the pointwise means is needed. The quantity B_k can grow with the number of action words. Uniformity in memory width is not uniformity in the size of the finite instrument being constructed.

10.2. Rationalization without a small-prefix denominator. Let G now be a finite-report acquisition of this form. For each θ , write

$$\pi_\theta(w) = \mathbb{P}_\theta(W = w), \quad p_\theta(w, h; y | a) = \mathbb{P}_\theta(W = w, Y_{1:k-1} = h_y, Y_k = y | a_{1:k-1} = h_a, a_k = a)$$

for its joint marked prefix masses under the indicated open-loop control word. Here $h = (h_y, h_a)$, and the conditioning on controls means intervention, not conditioning on the probability of choosing a control. Causality and the initial-mark convention imply

$$\sum_y p_\theta(w, h; y | a) = p_\theta(w, h)$$

independently of the current action a . A parameterized latent history-tree instrument may use these masses; its latent dependence on θ is part of the experiment, not information given to a simulator.

Suppose nonnegative rational arrays $t_\theta(w)$ and $t_\theta(w, h; y | a)$ approximate the indicated joint masses. Normalize $t_\theta(w)$ to obtain a rational initial-mark law. At each node (w, h, a) normalize its vector of children to obtain a rational report kernel. If a normalizing sum is zero, use a fixed rational probability vector, for example the uniform vector. This defines a coherent finite rational

marked causal experiment $G_{\mathbb{Q}}$. Its prefix probabilities need not equal the originally rounded joint masses; they are the probabilities generated by these normalized kernels. That distinction is essential.

Lemma 10.2 (Weighted normalization). *If $u, v \in [0, \infty)^b$, $s = \sum u_j$, and $\bar{u} = u/s$ when $s > 0$, then, for $\bar{v} = v/\sum v_j$ with an arbitrary probability vector chosen when its denominator vanishes,*

$$s\|\bar{u} - \bar{v}\|_{\text{TV}} \leq \sum_j |u_j - v_j|. \quad (10.2)$$

For $s = 0$ the left side is defined to be zero.

Proof. If both sums are positive, the triangle inequality gives

$$\sum_j |u_j - s\bar{v}_j| \leq \sum_j |u_j - v_j| + \left| \sum_j v_j - s \right| \leq 2 \sum_j |u_j - v_j|.$$

If $\sum v_j = 0$, the left side of (10.2) is at most $s = \sum |u_j - v_j|$. The remaining case is immediate. $\square$

Theorem 10.3 (Rational causal tree bridge). *The identity report channels compare G and $G_{\mathbb{Q}}$ in both directions, including the actual marks, with feedback error at most*

$$\rho = 1 \wedge \max_{\theta} \left[\sum_w |\pi_{\theta}(w) - t_{\theta}(w)| + \sum_{k=1}^T \sum_{w,h,a,y} |p_{\theta}(w, h; y | a) - t_{\theta}(w, h; y | a)| \right]. \quad (10.3)$$

Here the finite sum includes all open-loop histories at the specified time, not only the histories reached by a particular controller. The bound is independent of simulator and consumer memory. It remains valid with a visible selector and its joint reference law.

If every joint mass has computable rational enclosures of arbitrarily small width, the construction is effective and produces $\rho \rightarrow 0$. No lower bound for any nonzero prefix probability, and no decision whether a prefix has exactly zero probability, is required.

Proof. Couple the initial marks by their normalized laws. Lemma 10.2 bounds the initial failure probability by the first sum in (10.3). Use the same controller and selector on both sides. Conditional on a matched marked history, maximally couple the next reports. Under the original instrument, the probability of reaching a particular marked history under any deterministic controller is either zero or the corresponding open-loop prefix mass; for a randomized controller it is at most that mass. The probability of reaching it with the coupling still matched is smaller still. Thus its first-mismatch contribution is bounded by the original prefix mass times the conditional variation distance. Applying (10.2) at that node bounds this product by the sum of joint child-mass errors. Sum over all nodes and times. The coupling inequality gives the asserted joint variation distance. The estimate is symmetric as a comparison of the two emitted laws; identity channels require no state in either direction.

For effectiveness, take a nonnegative rational lower endpoint of a certified enclosure for every mass, clipping a negative lower endpoint at zero. If there are M mass entries for a parameter and every interval has width at most η/M , the displayed bound is at most η . Normalization uses only rational arithmetic. A zero rounded sum invokes the declared fallback. The weighted bound, rather than a bound on a ratio at an individual rare history, proves convergence. Conditioning on an independent visible selector leaves every bound unchanged. The mark is part of the finite tree and is never redrawn independently of its reports. $\square$

For the Gaussian acquisition its required masses are explicitly

$$\int_X p_\theta(x) \gamma(w \mid x) \prod_{j=1}^k \mathbb{P}\{m_{j,a_{1,j}}(x) + \sigma_j \xi_j \in I_{y_j}\} d\mu(x). \quad (10.4)$$

This is an integration problem for a specified preparation and trial family. The theorem does not infer its effective solvability from the mere words “standard Borel”. Rigorous enclosures for (10.4) are part of the effective data specification in Theorem 13.1; explicit physical families are treated in Section 14.

11. ENRICHED DEFECT CERTIFICATES AND THEIR CONVERGENCE

A small error of a Galerkin orbit need not force a chosen residual majorant to be small. We replace that implication by an estimator whose own convergence is proved. The approximation is allowed to be a finite tree of trial observables rather than powers of one compressed matrix. This enrichment changes the numerical presentation, not the physical experiment or the charged consumer register.

Let H be a separable real Hilbert space, $U(t)$ a strongly continuous orthogonal group with generator L , and C_a finitely many orthogonal operators. Write $T_a = C_a U(\Delta)$, with fixed rational $\Delta > 0$. Let $\mathcal{D}$ be a countable rational vector space in $D(L)$ whose real span is graph dense in $D(L)$, and suppose its rational span is also graph dense. It suffices to start with a countable graph-core family and all rational linear combinations.

Definition 11.1 (Effective graph data). The data are effective if vectors of $\mathcal{D}$ can be enumerated and rational intervals of any requested width can be computed for inner products among finitely many vectors of the forms

$$v, \quad Lv, \quad C_a v, \quad f_1, \dots, f_T, \quad v \in \mathcal{D}.$$

Only inner products actually used in the following norm expressions are necessary. The initial observables have specified H names. For physical probability certification we require, in addition, computable enclosures for (10.4) for finite trial families selected by this procedure and certified preparation bounds. These are finite analytic input operations; no deficiency optimizer or exact controlled orbit is an input operation.

For nodes $v_0, \dots, v_l \in \mathcal{D}$ at rational times $0 = s_0 < \dots < s_l = \Delta$, let $z(s)$ be their piecewise affine interpolant, and set

$$\begin{aligned} \mathcal{D}(z) &= \frac{1}{2} \sum_{j=0}^{l-1} (s_{j+1} - s_j) (\|d_j - Lv_j\| + \|d_j - Lv_{j+1}\|), \\ d_j &= (v_{j+1} - v_j) / (s_{j+1} - s_j). \end{aligned} \quad (11.1)$$

The interpolation lies in $D(L)$ at every time and is absolutely continuous in its graph norm. For a trial input $g \in \mathcal{D}$ and trial output $v \in \mathcal{D}$, define the one-step certificate

$$\mathcal{E}_a(g, z, v) = \|g - z(0)\| + \mathcal{D}(z) + \|C_a z(\Delta) - v\|. \quad (11.2)$$

No derivative of an intervened vector appears.

Lemma 11.2 (Certified path defect). *For every such triple,*

$$\|T_a g - v\| \leq \mathcal{E}_a(g, z, v).$$

Given $g \in \mathcal{D}$, an action a , and $\eta > 0$, there are rational nodes with $z(0) = g$ and a rational-core output v such that $\mathcal{E}_a(g, z, v) < \eta$.

Proof. On each interpolation interval, variation of constants gives

$$U(\Delta)z(0) - z(\Delta) = \int_0^\Delta U(\Delta - s)(Lz(s) - z'(s)) ds.$$

The identity follows first on each graph-differentiable interval and then by concatenation. The integrand's norm is bounded by the affine interpolation of its endpoint norms, since z' is constant on that interval and Lz is affine. Thus its integral is at most $\mathcal{D}(z)$. Orthogonality of C_a and the two endpoint errors give (11.2).

For existence put $u(s) = U(s)g$. It is continuously differentiable in H , continuous in the graph norm, and $u' = Lu$. Let a uniform rational mesh have size h . Its exact-node interpolant has derivative error bounded by the modulus of continuity of u' at scale h , and its L -image differs from $Lu(s)$ by at most the modulus of continuity of Lu at that scale. Both moduli tend to zero. Approximate each noninitial node by a member of $\mathcal{D}$ within h^2 in graph norm; keep the initial node equal to g . The change in the interpolant derivative is at most $2h$ and the change in its L -image at most h^2 . The endpoint norms in (11.1) obey these same modulus bounds, so $\mathcal{D}(z) \rightarrow 0$, not only the orbit error. Finally use density of $\mathcal{D}$ in H to approximate $C_a z(\Delta)$ by v . No assertion that $C_a z(\Delta) \in D(L)$ is needed. $\square$

Theorem 11.3 (Effective enriched controlled approximation). *Under the effective graph-data hypothesis there is an algorithm which, for a fixed finite horizon and rational tolerance $\eta > 0$, returns finite-core mean functions $\hat{m}_{k,w}$ and rational upper bounds $e_{k,w}$ such that*

$$\|T_{a_1} \cdots T_{a_k} f_k - \hat{m}_{k,w}\| \leq e_{k,w} < \eta, \quad w = (a_1, \dots, a_k), \quad 1 \leq k \leq T. \quad (11.3)$$

Each bound is a sum of an initial approximation error and certified one-step path defects. The algorithm terminates; no rate or polynomial complexity is asserted. Its certified physical marked error is

$$\alpha = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left(\sum_{w \in A^k} e_{k,w}^2 \right)^{1/2}, \quad (11.4)$$

and tends to zero along requested tolerances tending to zero. The source and approximation are compared by stateless identity report channels, including all independent exposed selectors.

Proof. Enumerate $\mathcal{D}$ and increasingly accurate rational norm upper bounds until a trial vector for f_k has error strictly below a prescribed small tolerance. Density ensures termination. For each step enumerate all finite rational time partitions and all rational core nodes and outputs, with initial node the trial input. Dovetail these candidates and interval precisions. Lemma 11.2 provides a candidate whose true defect is strictly smaller than the requested tolerance, so eventually its rational upper enclosure is also smaller. This is a terminating semidecision search with a proved strict witness. It does not test an undecidable equality of real numbers.

For each final acquisition k , build a suffix tree starting at the trial f_k . At the next level approximate T_a applied to each current trial vector using the preceding search. After k levels, the chronological word $a_1, \dots, a_k$ corresponds to applying a_k first to the observable and a_1 last. This gives exactly $T_{a_1} \cdots T_{a_k} f_k$, not its reversed physical dynamics. Orthogonality propagates the accumulated error additively. Allocate less than $\eta/(2(k+1))$ to the initial approximation and each step, and refine finite norm enclosures until their rational sum is below η . There are finitely many nodes and every search terminates.

All trial vectors used at finitely many nodes span a finite-dimensional subspace of H . Initialize its evaluation vector jointly with the actual mark and use the stored action-word table to read the appropriate linear functional at each time. This is a causal probability model. Its physical lookup table is not declared to be a finite consumer register, and its coefficient matrices are not called Markov matrices. The first-mismatch proof of Theorem 9.5, with the verified errors in (11.3), gives

(11.4). Every term tends to zero at fixed horizon. All numerical square roots and constants in an implementation are replaced by rational upper enclosures; their extra tolerances may also be made arbitrarily small. $\square$

Remark 11.4 (Existence, effectiveness, and the old Galerkin estimator). Graph-core density alone proves existence of arbitrarily small enriched defects. Effective graph data make the search an algorithm. Neither assertion says that the Galerkin majorants (9.12) converge along every graph-core sequence. Those estimates remain valid for every selected space and are kept in the article. The new estimator has a different convergence proof: its piecewise affine residual itself is made small. Effective integration of the resulting finite report probabilities is an additional operation, explicit in Definition 11.1, and is not inferred from Hilbert-space density.

12. EFFECTIVE GRAPH DATA FROM REGULAR COLLISION CHARTS

The effective-data hypothesis can be discharged for a class of genuine colliding systems; it need not remain an oracle for their exact flow. Here the particle number is fixed. No bound uniform in particle number, number of collisions, or kinetic scaling is used or concluded.

Fix $N \geq 2$, a flat square or cubic torus of positive rational side length, and a positive rational sphere diameter smaller than half that length. Assume the allowed configuration region has nonempty interior. Let μ be the normalized equilibrium law with independent standard Gaussian velocity coordinates, and let the finite intervention family consist of algebraic orthogonal transformations applied identically to every particle velocity. Flight times are rational. Noise scales are computable positive reals with supplied rational positive lower bounds. We use the classical almost-everywhere regular hard-sphere flow: on each bounded time interval almost every phase point has a finite itinerary of simple, nongrazing binary collisions, and the flow preserves equilibrium measure [6]. The following effective statements are consequences of that flow theorem and the explicit collision formulas, rather than additional collision-count estimates.

A bounded observable or preparation is called elementary-computable here if it has a supplied rational bound and admits uniformly computable rational interval enclosures on compact rational boxes of continuity, with mesh errors tending to zero. We allow piecewise descriptions whose exceptional boundaries are a specified finite union of nonzero polynomial zero sets. Globally computable continuous functions with known local moduli, and bounded piecewise polynomial functions, are examples. Preparations are nonnegative normalized densities of this class. A finite actual-mark channel has elementary-computable probabilities $\gamma(w \mid x)$ summing to one. These specifications exclude arbitrary noncomputable measurable inputs.

Lemma 12.1 (Effective full-measure chart exhaustion). *For any fixed finite collection of controlled words and bounded time windows, regular collision charts can be enumerated with rational box certificates. Their union has full equilibrium measure. Integrals of bounded functions which have computable interval enclosures on these charts have rational enclosures of arbitrarily small width, provided a uniform bound for the integrand is supplied. The assertion also holds with finitely many compact time-integration variables.*

Proof. On a velocity-cutoff box and a bounded time window, energy conservation gives a computable speed bound. Only finitely many periodic images can participate in a collision; enumerate all images within that displacement bound. The candidate time for a pair is a root of

$$|q_i - q_j + n\ell + t(v_i - v_j)|^2 = d_0^2,$$

where ℓ is the torus side and d_0 the diameter. On a simple nongrazing itinerary, the relevant discriminant is positive, the selected future root is separated from every competing root, and the normal relative velocity does not vanish. Reflection of the velocities is an explicit algebraic function of the positions and velocities at that root. An already realized contact has its zero

root removed; the outgoing sign specifies the next free flight. In unfolded coordinates there is no coordinate-wrap event to treat as a physical collision. A pair with zero relative velocity is handled by a strict positive separation bound on its whole flight interval, not by division by a vanishing quadratic coefficient. The same no-contact interval test handles sufficiently small perturbations when no root is present.

Enumerate finite collision lists, rational velocity and position boxes, and rational strict margins. Interval arithmetic for algebraic operations and square roots verifies the selected roots, their order, nongrazing margins, and absence of another collision before the final time. On sufficiently small boxes about any regular itinerary with a noncollision final time, all these strict inequalities hold. This is because there are finitely many collisions and every relevant gap and nongrazing angle is positive there. Thus enumeration produces a neighborhood of every such point. Endpoint-collision sets have measure zero. Reverse time uses the same construction. Algebraic kicks have computable inverses and preserve μ , so the finite intersections needed for controlled words and transformed chart inputs again have full measure. This proves effective enumerability and exhaustion; it does not assert a computable uniform collision-count bound.

The equilibrium measure itself is computable on boxes. The configuration normalizer is the volume of a bounded region defined by finitely many algebraic separation inequalities, with boundary of Lebesgue measure zero. Inside and outside rational-box approximations converge to its volume. A rational interior box provides a strictly positive lower bound for the normalizer; it can be found by enumeration because the interior is nonempty. Velocity-box masses are products of computable Gaussian interval probabilities. Gaussian tails provide effective cutoffs. These facts give rational upper and lower enclosures for μ and for the mass of every box used in the construction.

Replace finitely many accepted boxes by a disjoint dyadic refinement. Continue enumeration and refinement until the certified lower mass of their union exceeds $1 - \varepsilon$. This search terminates since the countable union has full measure. On the accepted compact boxes use interval enclosures and subdivision to bound the integrand's integral; on the complement use its known absolute bound M , at cost at most $M\varepsilon$. Refining the mass and integration enclosures gives any specified absolute error. Exceptional input-function boundaries are also null and can be excluded by strict interval tests for their nonzero defining polynomials. Compact time variables are handled by product-box integration; their collision-time sets are null by the finite-itinerary property and Fubini's theorem. At no point is an uncertified sample estimate for the omitted set used. $\square$

Let (ψ_j) be a countable uniformly described dense family of bounded, compact-velocity-supported continuous piecewise polynomial functions on phase space. For instance, use rational tent functions on successively finer coordinate boxes. Restrictions of this family are dense in $L^2(\mu)$. For rational $\delta > 0$ set

$$\kappa_\delta(s) = \frac{15}{16\delta}(1 - s^2/\delta^2)^2 \mathbf{1}_{[-\delta, \delta]}(s), \quad v_{j,\delta} = \int \kappa_\delta(s) U(s) \psi_j ds.$$

Let $\mathcal{D}$ be all rational linear combinations of these vectors.

Theorem 12.2 (Constructive microscopic graph data). *For the fixed-particle equilibrium system just specified, $\mathcal{D}$ is an enumerable graph core for L . Each vector has the exact representation*

$$Lv_{j,\delta} = - \int \kappa'_\delta(s) U(s) \psi_j ds. \tag{12.1}$$

All inner products required in Definition 11.1, for bounded elementary-computable acquisition observables, have arbitrarily accurate rational enclosures. The marked Gaussian prefix masses of every finite trial family in $\mathcal{D}$ also have such enclosures. Thus the complete effective-data hypothesis of Theorem 13.1 holds in this class, including its probability integration clause.

Proof. The kernel is continuously differentiable, compactly supported, has integral one, and vanishes at its support endpoints. The strong group identity and integration by parts prove domain membership and (12.1). If $u \in D(L)$, its mollified orbit converges to u in graph norm as $\delta \downarrow 0$. At fixed δ , approximating u in H by rational combinations of the ψ_j approximates its mollification in graph norm, since

$$\|A_\delta(u - \psi)\| + \|LA_\delta(u - \psi)\| \leq (\|\kappa_\delta\|_1 + \|\kappa'_\delta\|_1)\|u - \psi\|.$$

A diagonal choice proves graph density. No decision of linear independence between formal core vectors is needed by the enriched search.

For inner products insert the time-integral representations on each side and use Fubini. The resulting integrands are bounded products of known kernels and of observables evaluated along finitely many exact collision itineraries, possibly after an algebraic kick. Known sup-norm bounds for ψ_j , κ_δ and κ'_δ bound these products. Lemma 12.1 therefore computes their integrals, including cross terms involving $C_a v$ and the given bounded f_k . The signs of κ' present no difficulty because absolute bounds are available.

For completeness, the time-integral value of a core function itself admits uniform enclosures on each compact regular itinerary chart. Enclose all collision times there. Away from small strips containing those times, the algebraic flow formulas and input observables have uniform interval approximations. The strips contribute at most their total length times the known kernel and observable bounds. Strict itinerary margins allow these strips to shrink under subdivision. Thus both $v(x)$ and $Lv(x)$ are computable continuous functions on that chart, even if a velocity-dependent ψ_j jumps at a collision. The integral across its moving jump is controlled by the strip bound, not by pretending the jump is absent.

Every trial mean is a finite rational combination of such bounded core functions. Gaussian cell probabilities are computable continuous functions of that mean, including the two tails, and their derivatives are bounded by the positive noise scales. The integrand in (10.4), multiplied by a bounded preparation and actual-mark probability, consequently satisfies the hypotheses of Lemma 12.1. The lemma computes its mass enclosures. It also supplies L^2 preparation and trial-norm enclosures. This verifies all finite operations required by Definition 11.1. The exhaustive searches may have no useful uniform rate, but each requested enclosure terminates by the full-measure exhaustion proved above. $\square$

Corollary 12.3 (Certified approximation for a colliding physical class). *For two systems in Theorem 12.2 with a common finite control interface, finite parameter and mark alphabets, bounded elementary-computable preparations and observables, rational positive noise variances and rational observation times, the procedure of Theorem 13.1 produces convergent same-budget intervals for their physical intrinsic deficiency. It uses certified collision charts rather than an exact-flow integration oracle. It does not require the acquisition observables to be conserved quantities.*

Proof. The theorem supplies the effective graph data and marked prefix integrals; Theorem 11.3 supplies convergent analytic certificates; the finite report and rational-tree bridges preserve the resource signature. Apply Theorem 13.1. One may work in the full labeled L^2 space. For symmetric input data the construction also restricts to its symmetric subspace by symmetrizing the tent family; the flow and identical-particle kicks commute with that finite symmetrization. $\square$

The use of a null exceptional set is constructive here because its regular complement has an effective increasing chart enumeration and because the probability of each accepted finite cover is certified. A bare assertion of almost-everywhere computability without those two properties would not justify the conclusion. Conversely, the argument is an exhaustion algorithm, not a practical complexity bound or a uniform estimate in the Boltzmann–Grad regime.

13. PHYSICAL INTRINSIC RESOURCE CERTIFICATION

We can now state a single conclusion joining the analytic and finite parts of the theory. The original experiments, not their approximations, are the objects whose intrinsic deficiency is bounded. Fix one of the private, hidden-selector, or visible-selector signatures, denoted b ; its state widths and, where relevant, selector cardinality are fixed. Write $\delta_b(E, F)$ for the corresponding intrinsic deficiency, with source E simulating target F . The marked feedback comparison always uses the actual initial mark. For a visible selector its reference law is the joint law specified in Section 2.

Theorem 13.1 (Same-budget physical certification). *Let E, F be finite-horizon active Gaussian acquisitions on Hilbert probability models of the form considered above, with finite parameter, action and actual-mark alphabets, positive noise variances, and certified L^2 preparation bounds. For each model choose a finite trial presentation with rigorous marked error α_E, α_F , supplied either by Theorem 9.5 or by Theorem 11.3. Choose finite report channels as in Lemma 10.1 and rational marked trees as in Theorem 10.3, with certified errors β_E, β_F and ρ_E, ρ_F . Put*

$$a = 1 \wedge (\alpha_E + \beta_E + \rho_E), \quad c = 1 \wedge (\alpha_F + \beta_F + \rho_F).$$

Let $[L, U]$ be any verified local-certificate interval for $\delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})$ at the original signature b . Then the original physical deficiency belongs to

$$[\max\{0, L - a - c\}, \min\{1, U + a + c\}]. \quad (13.1)$$

An upper witness is an executable simulator of the original physical experiment with exactly the declared persistent-state and selector budget. A strict lower endpoint is an impossibility certificate at that same budget; it does not refer to a convexified private feasible set.

For the effective assertion, the noise scales are computable reals with supplied rational positive lower bounds. Under effective graph data, including the marked cell-integral operation in Definition 11.1, there is a terminating procedure producing intervals of the form (13.1) of arbitrarily small prescribed width. Their nested intersections converge to $\delta_b(E, F)$. The procedure uses only finite rational certificates and certified analytic input operations. It does not assume attainment of the original nonfinite optimization.

Proof. Each of the three presentation changes is implemented by a stateless, seed-free, parameter-blind and task-blind report channel. The physical trial change uses the identity report channel; the quantization change uses Q in one direction and transiently randomized H in the other; the rationalization change uses the identity finite-report channel. Their compositions give two-sided marked feedback comparisons $E \leftrightarrow E_{\mathbb{Q}}$ with error a and $F \leftrightarrow F_{\mathbb{Q}}$ with error c . Fresh per-report randomness in a presentation channel is integrated into a simulator row; it is not a persistent public or hidden mode. All comparisons include the joint reference of a visible selector.

For an executable finite upper witness, first pass a report of E through the forward presentation, apply the finite simulator, and pass its output through the reverse presentation for F . The stored state and persistent selector are exactly those of the finite witness. The marked feedback triangle inequality bounds its error by $U + a + c$. Conversely, any physical simulator with error smaller than $L - a - c$ could be composed with the opposite stateless presentations to give a finite simulator of the same signature with error smaller than L . This contradicts the verified finite lower certificate. Equivalently, Theorem 5.6 gives

$$|\delta_b(E, F) - \delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})| \leq a + c.$$

Taking infima in this argument requires no nonfinite minimizer.

For effectiveness, first choose enriched tolerances so that the two rational upper bounds for α are as small as requested. The terminating search of Theorem 11.3 supplies the trial means and their norm bounds. Choose rational R large and h small until the explicit rational upper bounds for β are small. There are then finitely many marked prefix masses. Request their enclosures at

a precision making the finite sum in (10.3) small, and normalize rational lower endpoints. This gives actual finite rational instruments, not floating-point transition tables.

For these fixed finite instruments, Theorem 5.4 supplies convergent dyadic local lower certificates and executable rational upper witnesses. Refine until $U - L$ is below the requested remaining budget; convergence, or the explicit causal oscillation modulus, ensures termination. Although the finite alphabet and row dimension may grow as the physical tolerance decreases, at each stage this is a finite problem and the mesh is selected for that problem. Thus the total interval width is at most $U - L + 2(a + c)$ and can be made arbitrarily small. Intersect each interval with its predecessors. They all contain the same true value, so the intersections are nonempty and nested, and their lengths tend to zero. $\square$

Remark 13.2 (What is constructive in the theorem). The quantitative implication (13.1) holds for any rigorously supplied analytic enclosures. Its effective-convergence assertion has an explicit analytic data hypothesis. A graph core known only to exist is not, by itself, a numerical integration algorithm. Neither arbitrary real preparations nor arbitrary measurable marks are silently declared computable. Section 14 gives genuine hard-sphere data for which the required calculations can be certified, including an exact continuous-report deficiency. The private certificate search remains nonconvex and may be expensive; the witness-dimension theorem reduces the number of distinct tests, not the number of cells or the cost of real quantifier elimination.

Corollary 13.3 (Regenerative risk certificates). *Retain the actual synchronous reset and stationary-policy signature of Section 8. For losses in $[0, 1]$, certified marked presentation errors ϵ_j for ages $0 \leq j \leq N$ give a uniform stationary-risk error at most*

$$\eta \sum_{j=0}^N (1 - \eta)^j \epsilon_j + (1 - \eta)^{N+1}.$$

The same bound holds for each lower support function of the stationary risk body. Finite-age rational local certificates may be transferred at their declared budgets by Theorem 13.1. Under its effective hypotheses the presentation part of the displayed risk bound can be made arbitrarily small, and then N can be increased.

Proof. Condition the stationary regenerative law on its age. The law of age j has weight $\eta(1 - \eta)^j$, and the omitted ages have total weight $(1 - \eta)^{N+1}$. Apply the marked bounded-loss inequality to each retained age and then sum. The bound precedes optimization, so it also bounds lower support functions. Finite-age certificates concern those finite-age experiments; this corollary does not identify a finite-age optimized deficiency with an unrestricted infinite-history deficiency. The register is actually cleared at reset and any persistent selector remains charged, exactly as in the inherited theorem. $\square$

14. CERTIFIED DATA FOR ACTIVE HARD SPHERES

The preceding theorems require analytic data, not only matrix regressions. We give two explicit families. The first has an exactly computable continuous-report intrinsic deficiency. The second has a nonzero graph residual obtained from a collision-compatible position observable. The hard-sphere flow and its almost-everywhere realization are the same as in the inherited active microscopic construction; no new assumption on a kicked vector's generator domain is introduced.

14.1. A conserved mode with an exact marked deficiency. Take $N \geq 2$ identical hard spheres on a flat torus with a nonempty configuration region. Normalize the equilibrium law to have independent standard Gaussian velocity coordinates. Its configuration factor is uniform on

the allowed configurations. For a fixed coordinate put

$$X = N^{-1/2} \sum_{i=1}^N v_{i,1}, \quad W = \operatorname{sgn} X.$$

Then X is standard Gaussian and W is a fair actual initial mark. Elastic collisions preserve total momentum, hence $U(t)X = X$ and $X \in \bigcap_{j \geq 1} D(L^j)$ with $LX = 0$. The actions $a \in \{-1, 1\}$ reverse all velocities or leave them fixed, so $\bar{C}_a X = aX$. They are physical interventions, even though the selected observable lies in a conserved sector. In the normalized space $V = \operatorname{span}\{X\}$, the exact Gram data are

$$G = [1], \quad J = [0], \quad B_a = [a], \quad R = [0], \quad S_a = [0]. \quad (14.1)$$

All entries are rational. These identities hold for the full colliding system, not a collision-free restriction of its equilibrium law.

Consider one acquisition after the chosen action and any fixed flight time. Source E reports a fresh standard Gaussian independent of the state, while target F reports

$$Y = aX + \xi, \quad \xi \sim N(0, 1).$$

There is no informative earlier report. Source and target carry the same actual mark W . The simulator knows the current action, not X or W ; every persistent selector is independent of the physical state.

Theorem 14.1 (An exact continuous physical certificate). *For every nonempty finite private width profile, and every finite hidden or visible independent-selector budget,*

$$\delta_b(E, F) = \frac{1}{4}. \quad (14.2)$$

The upper witness is the stateless simulator that emits an independent $N(0, 2)$ report. A lower witness is the marked event $\{W = \operatorname{sgn}(aY)\}$. The report-sign coarsening of F has the exact rational joint table

$$\mathbb{P}_F(W = w, Z = z \mid a) = \begin{cases} 3/8, & z = aw, \\ 1/8, & z = -aw, \end{cases} \quad Z = \operatorname{sgn} Y. \quad (14.3)$$

It also has deficiency $1/4$ from a source independent of W .

Proof. For any source simulator its emitted report is independent of W , conditional on an exposed selector and the externally fixed action. The indicated event therefore has probability $1/2$. At the target, $aY = X + a\xi$. The plane vector $(X, a\xi)$ is rotationally invariant; the two sign-disagreement wedges between X and $X + a\xi$ have total angle $\pi/2$. Consequently agreement has probability $3/4$ and the variation discrepancy is at least $1/4$. The action may be fixed in this test, so taking the feedback supremum cannot weaken the bound. This argument holds in each exposed-selector fibre and also after hiding a selector.

For the upper witness write $\varphi_{\sqrt{2}}$ for the $N(0, 2)$ density. The conditional Gaussian law $X \mid Y = y$ has mean $ay/2$ and variance $1/2$. Thus the target joint densities with respect to counting measure on w and Lebesgue measure on y are

$$p(w, y) = \varphi_{\sqrt{2}}(y) \Phi(way/\sqrt{2}).$$

The source witness has joint density $q(w, y) = \frac{1}{2} \varphi_{\sqrt{2}}(y)$. The sign of $p - q$ is positive exactly on $w = \operatorname{sgn}(ay)$ and negative on its complement. Hence that event attains the total variation, which is $3/4 - 1/2 = 1/4$. There are only the two possible actions; the same calculation holds for both, and for mixtures determined by any independent selector. The witness uses no persistent state. Integrating over the two report signs and using symmetry proves (14.3). Its variation

discrepancy from the independent fair-sign witness is again $1/4$, and the same event proves the lower bound. $\square$

The sign table is an exact finite lower-witness presentation, not a two-sided zero-error reconstruction of the continuous Gaussian report. The continuous upper bound above is proved directly. For an arbitrary small two-sided bridge one must instead use the partitions and error budgets in Section 10. Confusing these two uses of a coarsening would give an incorrect certificate.

Proposition 14.2 (Effective cell data in the conserved sector). *For finitely many repeated acquisitions of X , with the two actions above, rational noise variances bounded away from zero, and the mark $\text{sgn } X$, every marked Gaussian prefix mass in (10.4) admits rational enclosures of arbitrarily small prescribed width. The exact trial space has zero graph and intervention errors for every horizon. Consequently Theorem 13.1 gives a convergent same-budget finite rational certificate procedure for this active physical family.*

Proof. Every controlled mean is X or $-X$. The mass is a one-dimensional integral on one of the half-lines of the standard Gaussian density times finitely many Gaussian interval probabilities. Its tail outside $[-M, M]$ is at most M^{-2} , by the second moment. On the remaining compact interval, each interval probability and its derivative has a computable uniform bound: its derivative is a difference of two Gaussian densities divided by its positive noise scale. Products therefore have a computable Lipschitz bound. Uniform Riemann sums have an explicit mesh error given by this bound times the interval length and mesh size. Gaussian density and distribution values at algebraic arguments have rational enclosures: integrate the power series of $\exp(-u^2/2)$ on a bounded interval with a Taylor remainder bound, using symmetry and a Gaussian tail bound for unbounded endpoints. Square roots and the normalizing constant have elementary rational interval algorithms. Combining these enclosures with the mesh and tail errors gives the requested rational mass interval. All quantities are scalar; the configuration-volume normalizer cancels. Equation (14.1) removes the analytic trial error for every controlled word. Apply the finite-report and rational-tree constructions and then the finite local-certificate theorem. This proves an actual procedure for this family, not computability of arbitrary preparations. $\square$

14.2. A nonzero collision-compatible graph residual. Consider two spheres of diameter one on the two-dimensional square torus of side ten. The allowed ordered configuration volume is $100(100 - \pi)$; velocities remain independent standard Gaussians. Let $b(u) = (1 - u^2)^2$ on $|u| \leq 1$ and zero elsewhere, and let $h = 1/4$. Place squares of half-side h at $c_1 = (2, 5)$ and $c_2 = (8, 5)$. Let $\chi_1(q_1, q_2)$ be the product of the four rescaled functions b when particle one is near c_1 and particle two near c_2 , and put $f = \chi_1(q_1, q_2) + \chi_1(q_2, q_1)$. The two supports are disjoint, are separated from the collision boundary, and define a symmetric C^1 position observable on the torus.

Proposition 14.3 (Exact nonzero physical Gram data). *The vector f belongs to $D(L)$ and $Lf = v_1 \cdot \nabla_{q_1} f + v_2 \cdot \nabla_{q_2} f$. With $V = \text{span}\{\phi\}$, $\phi = f/\|f\|_2$, its exact data are*

$$J = 0, \quad R = [192], \quad B_{\pm} = 1, \quad S_{\pm} = 0, \quad \|f\|_2^2 = \frac{2h^4(256/315)^4}{100(100 - \pi)}. \quad (14.4)$$

Here the actions still reverse all velocities or leave them fixed. For one acquisition of f with unit Gaussian noise after $\Delta = 1/1000$, under equilibrium preparation and with any initial finite mark channel, the static trial mean f and the true active physical acquisition have two-sided marked feedback error less than $1/100000$. All constants in this last assertion follow from rational Gram enclosures and $3 < \pi < 4$.

Proof. The position support is at positive distance from contact, including periodic images; its minimal separation is greater than one. On a regular orbit f is continuously differentiable between collisions, and its transport derivative vanishes near each collision configuration. Thus the integral

transport identity holds across collisions without a jump. Its difference quotients are bounded in absolute value by a constant times the conserved total speed. This is square integrable under the Gaussian velocity law. Dominated convergence gives the strong L^2 generator derivative claimed. This also proves the required collision-domain membership rather than presuming a smooth-boundary extension of a kicked function.

Direct polynomial integration gives

$$I_0 = \int_{-1}^1 b(u)^2 du = 256/315, \quad I_1 = \int_{-1}^1 b'(u)^2 du = 256/105.$$

The two disjoint product supports give $\|f\|^2 = 2h^4 I_0^4/Z$, $Z = 100(100 - \pi)$. Independence, centering and unit variance of velocity coordinates give

$$\langle f, Lf \rangle = 0, \quad \|Lf\|^2 = \frac{4I_1}{h^2 I_0} \|f\|^2 = 192 \|f\|^2.$$

Velocity reversal fixes the position vector f , so $B_{\pm} = 1$ and $S_{\pm} = 0$. The displayed Gram residual follows. Since $f \in V$, there is no projection error and the one-step residual is $\Delta\sqrt{192}\|f\|$. The two-action bound in Theorem 9.5 is therefore

$$\alpha \leq \frac{\Delta\sqrt{384}\|f\|}{\sqrt{2\pi}}.$$

Using $I_0 < 1$, $h = 1/4$ and $Z > 9600$ gives $\|f\|^2 < 1/1228800$. Together with $2\pi > 6$ this yields $\alpha^2 < 64/(10^6 \cdot 1228800) < 10^{-10}$. Every inequality is rational except the explicitly enclosed π ; no floating-point Gram estimate is used. A mark coupled to the original state stays coupled in the first-mismatch argument. $\square$

These two examples answer different questions. A conserved sector can carry nontrivial hidden information and an exact positive deficiency; a localized transport sector supplies genuinely nonzero graph residuals and certified error bounds. The velocity interventions change the physical dynamics in both cases. Neither example is a proof of the particle-number-uniform nonlinear kinetic corrector in the historical B4 program. Their downstream edge is the explicit marked physical-to-finite certification theorem, with its own stated analytic inputs and error topology.

15. A NONCONSERVED COLLISION-COMPATIBLE ACQUISITION

The following example lies on the full equilibrium phase space of an interacting hard-sphere system. It is neither a collision-free restriction nor an observable constant along the flow. Its explicitly solvable collective mode permits simultaneous evaluation of a nonzero generator residual and of a marked decision deficiency. The orbit factors through total momentum, so collisions cancel from this particular observable. The residual measures projection error, not interaction strength. The collision-sensitive comparison is proved in Section 4; the solvable collective calculation is retained here as a separate exact benchmark.

Fix $N \geq 2$ equal-mass hard spheres on the square flat torus of side $\ell > 0$, with positive admissible configuration volume. Velocities have independent standard Gaussian coordinates under equilibrium μ . Let $U(t)$ be the full hard-sphere Koopman group, and allow the two physical interventions $v \mapsto av$, $a \in \{-1, 1\}$, before each flight of length Δ . Define the circle-valued coordinate and the normalized total momentum

$$\vartheta = \frac{2\pi}{\ell} \sum_{i=1}^N q_{i,1} \pmod{2\pi}, \quad X = N^{-1/2} \sum_{i=1}^N v_{i,1}, \quad \omega = \frac{2\pi\sqrt{N}}{\ell}, \quad f = \cos \vartheta.$$

The exponential of the displayed coordinate is well defined on the torus; no choice of lifted particle positions enters f .

Theorem 15.1 (Collective domain and controlled-word reduction). *The coordinate ϑ is Haar uniform and independent of the standard normal X . On the full-measure domain of the hard-sphere flow,*

$$U(t)f = \cos(\vartheta + \omega Xt), \quad L^j f = \omega^j X^j \cos(\vartheta + j\pi/2), \quad \|L^j f\|_2^2 = \frac{\omega^{2j}(2j-1)!!}{2}. \quad (15.1)$$

Here $f \in \bigcap_{j \geq 1} D(L^j)$ and $(-1)!! = 1$. In the normalized one-dimensional trial space spanned by $\varphi = \sqrt{2}f$, the matrices of the preceding residual theorem are

$$G = 1, \quad J = 0, \quad R = \omega^2 > 0, \quad B_{-1} = B_1 = 1, \quad S_{-1} = S_1 = 0.$$

For a word $w = (a_1, \dots, a_k)$ put

$$s(w) = \Delta \sum_{j=1}^k \prod_{i=1}^j a_i.$$

Then $T_{a_1} \cdots T_{a_k} f = \cos(\vartheta + \omega X s(w))$. Exactly $k+1$ distinct observable functions occur among the 2^k words, indexed by $s/\Delta \in \{-k, -k+2, \dots, k\}$.

Proof. Simultaneous translation of every particle by r in the first torus coordinate preserves the admissible configuration measure and shifts ϑ by $2\pi Nr/\ell$. These shifts exhaust the circle, so its push-forward measure is Haar. Configuration and velocity are independent, and Gaussian addition proves the assertion about X . Elastic collisions conserve total momentum and do not jump particle positions. Between collisions the derivative of ϑ is ωX . This proves the orbit formula, including the periodic identifications, on the classical almost-everywhere flow domain. Difference quotients of that formula, dominated successively by polynomials in $|X|$, converge in $L^2(\mu)$ since all Gaussian moments are finite. Induction gives membership in every generator power domain and the derivative formula. Independence and the uniform circle integral of squared sine or cosine give the norm formula. This also establishes collision compatibility in the actual generator domain, rather than by applying a formal derivative to arbitrary boundary data.

The inner product of $\cos \vartheta$ and $X \sin \vartheta$ vanishes. Thus $J = 0$ and the squared residual norm of $L\varphi$ is ω^2 . A velocity reversal fixes φ , giving $B_a = 1$ and $S_a = 0$. After the j th intervention the total momentum is $(a_1 \cdots a_j)X$, so summing flight displacements proves the word formula in the order $T_a = C_a U(\Delta)$. Every sequence of k signs can be the successive cumulative products, which gives all $k+1$ clock values. They are distinct as observable functions because

$$\|\cos(\vartheta + \omega X s) - \cos(\vartheta + \omega X t)\|_2^2 = 1 - \exp\{-\omega^2(s-t)^2/2\} > 0 \quad (s \neq t).$$

□

Theorem 15.2 (Explicit converging finite physical presentations). *Let $m \geq 1$ and replace the mean at clock s by*

$$f_{m,s} = \sum_{j=0}^{m-1} \frac{s^j}{j!} L^j f.$$

For $|s| \leq k\Delta$ the mean error is bounded by

$$E_{m,k} = \frac{(k\Delta\omega)^m}{m!} \sqrt{\frac{(2m-1)!!}{2}}. \quad (15.2)$$

For Gaussian report noises $\sigma_k > 0$ and preparations satisfying $\sup_\theta \|p_\theta\|_2 \leq D$, the exact and polynomial acquisitions have two-sided stateless marked feedback comparison error at most

$$\alpha_{m,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{\sqrt{k+1}}{\sigma_k} E_{m,k}. \quad (15.3)$$

The same actual initial mark and every exposed independent selector are preserved. The estimate is uniform over all consumer register widths, and $\alpha_{m,T} \rightarrow 0$ at every fixed finite horizon. The physical trial functions and all their generator and intervention Gram entries are explicitly integrable against equilibrium.

Proof. The integral Taylor remainder for the unitary orbit on $D(L^m)$ has norm at most $|s|^m \|L^m f\|_2 / m!$, also for negative s . Theorem 15.1 gives (15.2). Its successive ratio is

$$\frac{E_{m+1,k}}{E_{m,k}} = k\Delta\omega \frac{\sqrt{2m+1}}{m+1} \rightarrow 0,$$

which is an effective bound for the convergence claimed here.

Couple the two acquisitions from the same initial x, W and the same independent selector. Until the first unequal report, their controls agree. Two unit-dimensional Gaussians with the same variance have variation distance at most their mean difference divided by $\sigma_k \sqrt{2\pi}$; this follows by integrating the derivative of the translated Gaussian density. At fixed x , the probabilities of all matched prefixes sum to at most one. Keep those probabilities until bounding the mismatch by the largest mean error at that x . There are only $k+1$ different errors at time k , even though there are 2^k words. Cauchy–Schwarz gives

$$\int p_\theta(x) \max_s |e_{m,s}(x)| d\mu(x) \leq D \left(\sum_s \|e_{m,s}\|_2^2 \right)^{1/2} \leq D\sqrt{k+1} E_{m,k}.$$

Sum first-mismatch probabilities over k . The construction in both directions uses the identity report interface, has no added persistent state, and couples the actual mark, proving (15.3). No bound on feedback-conditioned preparation densities was used.

The trial family consists of $X^j \cos \vartheta$ and $X^j \sin \vartheta$ for $j < m$. Its inner products are products of circle integrals and Gaussian moments. Generator entries multiply by ωX and rotate sine and cosine; the interventions replace X by $-X$. Every required finite entry is thus a polynomial in ω with rational coefficients before normalization. This supplies exact symbolic expressions and arbitrarily tight rational interval enclosures. The polynomial means may be unbounded in X ; that is harmless for (15.3). For finite-report integration one first truncates $|X|$ with its explicit Gaussian tail, then integrates bounded Gaussian report probabilities over the resulting compact cylinder. For the equilibrium and mark used below, the only mark boundaries are $\vartheta = \pi/2, 3\pi/2$, and can be split explicitly. On each piece trigonometric, polynomial and Gaussian distribution functions have computable uniform moduli. Rational upper and lower Riemann enclosures therefore converge. This verifies the effective marked integration hypothesis of the finite-presentation theorem for this class, without an exact hard-sphere flow integration oracle. $\square$

Remark 15.3. The signed clock is an index of physical mean functions, not an unpriced extra register offered to a consumer. The width-uniform proof uses equality of mean functions to reduce a bound; it does not allow a finite-state simulator to remember its entire control word. Neither the polynomial coefficient vector nor its real dimension is a finite consumer-label count. The underlying microscopic system still has collisions; solvability here comes from the conserved total momentum that transports a *nonconserved* position observable.

15.1. One actual mark, one nonzero residual, and a positive certificate. Let $W = \text{sgn} \cos \vartheta$ in the initial equilibrium state; the zero set has measure zero. Both experiments have a singleton initial report and allow $a \in \{-1, 1\}$ before their single acquisition. The source reports an independent standard Gaussian. The target reports

$$Y = \cos(\vartheta + a\tau X) + \xi, \quad \tau = \omega\Delta, \quad \xi \sim N(0, 1)$$

with noise independent of the actual initial mark and microscopic state. Let Φ and ϕ denote the standard normal distribution and density.

Theorem 15.4 (Exact marked physical deficiency). *Here the action precedes any informative report and its reversal symmetry leaves the relevant marked law unchanged. In the visible comparison the independent selector is retained jointly with the same selector law on the target side. For every nonempty finite private, hidden-selector, or visible-selector signature in this one-acquisition model, its intrinsic deficiency equals*

$$d(\tau) = \mathbb{E}[W\{\Phi(\cos(\vartheta + \tau X)) - 1/2\}] > 0. \quad (15.4)$$

It is attained by the stateless simulator that draws the target's unconditional report marginal. If

$$c_k(\tau) = 2^{-(2k+1)} \sum_{j=0}^{2k+1} \binom{2k+1}{j} \frac{2(-1)^{(|2k+1-2j|-1)/2}}{\pi|2k+1-2j|} e^{-(2k+1-2j)^2\tau^2/2},$$

then

$$d(\tau) = \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k c_k(\tau)}{2^k k! (2k+1)}. \quad (15.5)$$

The error after $k = K$ is at most $[\sqrt{2\pi} 2^{K+1} (K+1)! (2K+3)]^{-1}$.

Proof. The action is chosen before any informative report. Reversal of the independent symmetric X does not change the joint law of (W, Y) , including conditionally on any exposed independent seed. Set $H_\tau(\theta) = \mathbb{E} \operatorname{sgn} \cos(\theta + \tau X)$. This is the circle heat evolution of $\operatorname{sgn} \cos \theta$ at time $\tau^2/2$. It is strictly positive on $(-\pi/2, \pi/2)$ and strictly negative on the opposite arc. To see the sign without assuming it from a Fourier expansion, reflect the heat solution oddly across both endpoints of the positive arc. Its restriction solves the heat equation on that interval with zero boundary values and initial value one. Equivalently it is the survival probability of Brownian motion killed at those endpoints. This is positive in the interior at every finite positive time. The odd reflected solution agrees with the circle solution by uniqueness (the difference has zero initial data and its squared integral has nonpositive derivative). At $\tau = 0$ the same sign assertion is immediate.

The final angle is uniform. Reversibility of the Gaussian circle convolution gives $\mathbb{E}[W \mid \vartheta + \tau X = \theta] = H_\tau(\theta)$. Let ν be the unconditional density of Y . The density of (W, Y) , minus $\frac{1}{2}\nu(y)$ in its w th row, is

$$\frac{w}{4\pi} \int_0^{2\pi} H_\tau(\theta) \phi(y - \cos \theta) d\theta.$$

Pair points θ and $\theta + \pi$. On the positive arc both $H_\tau(\theta)$ and $\cos \theta$ are positive, and $\phi(y - \cos \theta) - \phi(y + \cos \theta)$ has the sign of y . Thus the density difference has the sign of wy , and is not zero. Its positive event is $\{wy > 0\}$. The excess probability of that event is (15.4), proving positivity and giving a stateless upper witness of that variation distance.

Every source simulator at the declared signature, conditionally on its exposed seed, produces a report independent of the fair W . For any such report law the event $\{wy > 0\}$ has probability at most $1/2$, including possible mass at zero. The target probability is $1/2 + d(\tau)$. This supplies the matching lower bound for private and hidden randomization; the same bound in each visible fiber supplies it for visible randomization. Mixtures of the two action choices do not change this conclusion.

Finally, uniformly for $|z| \leq 1$,

$$\Phi(z) - 1/2 = \frac{1}{\sqrt{2\pi}} \sum_{k \geq 0} \frac{(-1)^k z^{2k+1}}{2^k k! (2k+1)}.$$

The absolute remainder is bounded by the next term, by the alternating Taylor expansion of $e^{-u^2/2}$ followed by integration. Expand $\cos^{2k+1}$ into its finite complex exponential sum. For odd n ,

$$\mathbb{E}[W \cos(n\vartheta)] = \frac{2(-1)^{(|n|-1)/2}}{\pi|n|}, \quad \mathbb{E}e^{in\tau X} = e^{-n^2\tau^2/2}.$$

The sine terms vanish. These identities give c_k and the claimed series and remainder, with no unproved exchange of a nonuniform sum. $\square$

Corollary 15.5 (A rational, nonvacuous collision certificate). *For $N = 2$, $\ell = 10$, any rational diameter in $(0, 5)$ giving nonempty interior, and $\Delta = 1/2$, the same physical acquisition satisfies*

$$\frac{205809122887}{10^{12}} < d(\tau) < \frac{205809122889}{10^{12}}, \quad R = \frac{2\pi^2}{25} > 0.$$

The degree-fifteen physical Taylor presentation has marked comparison error less than $3 \cdot 10^{-11}$ for the single report.

Proof. Here $\tau^2/2 = \pi^2/100$. Evaluate (15.5) through $K = 10$ with outward rational arithmetic and add its stated remainder. For reproducibility the source supplies every arithmetic operation: $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ is bounded by alternating series; square roots are enclosed by rational bisection; exponentials are range-reduced to arguments in $[0, 1]$, enclosed by alternating series, and squared back with outward rounding. The resulting enclosure has lower endpoint greater than the displayed lower rational and upper endpoint smaller than the displayed upper rational. All comparisons are integer comparisons. Applying (15.2) with $m = 16$, and replacing $\sqrt{2}$ in (15.3) by the larger rational 2, yields the final strict bound. The generator residual follows exactly from Theorem 15.1. The supplied checker redoes these finite rational computations; the analytic justification is the preceding domain, coupling and series proofs, not the execution of that checker. $\square$

16. A JOINT PHYSICAL AND NONCONVEX CERTIFICATE THEOREM

The finite testing and microscopic approximation results now give a single lower–upper certificate for the original resource problem. The stateless transfer inequality and physical-to-rational bridge are those of Theorem 13.1; the polynomial lower module is a separate strengthening, not a new physical transfer principle. Unlike a constant mixture of tests, the polynomial lower certificate can separate a target from a private behavior image even when the target belongs to its convex hull.

Theorem 16.1 (Constructive same-budget physical testing). *Fix a finite horizon and one of the private, hidden-selector, or visible-selector signatures. Let E, F be marked physical acquisitions with finite compatible controls and marks. Suppose that rational finite presentations E_j, F_j have executable two-sided stateless marked feedback errors $a_j, c_j \rightarrow 0$, with computable upper bounds. For the algorithmic assertion require the uniform enumeration and rational error enclosures of Proposition 3.6, not merely individual computability of each presentation. Let $P_{p,L}$ be the polynomial (6.2) of the finite intrinsic deficiency problem for E_j, F_j , and let a nonnegative rational Bernstein array certify it. Let an actual rational simulator table at the same signature have finite error U . Then*

$$\max\{0, L - a_j - c_j\} \leq \delta_b(E, F) \leq \min\{1, U + a_j + c_j\}. \quad (16.1)$$

There is a terminating dovetailed search for such an interval of any prescribed positive rational width. Every lower certificate is checked by a finite polynomial identity and sign comparisons; every upper certificate is an executable table at the original budget. The visible case retains the selector in both compared marked laws.

The hypotheses hold for the effective fixed-particle collision class of Theorem 12.2, and in particular for the nonconserved collective acquisitions of Theorem 15.2 with equilibrium preparation and a finite computable marked channel with the piecewise effective integration property proved there. For the sign mark of Theorem 15.4, that integration property is explicit.

Proof. The finite polynomial identity gives $L \leq \delta_b(E_j, F_j)$ by Theorem 6.1; direct evaluation of the rational table gives the reverse bound by U . Compose it with the two stateless physical

presentation channels. Their identity state costs preserve all retained widths and selector budgets. The marked triangle and presentation-stability inequalities give

$$|\delta_b(E, F) - \delta_b(E_j, F_j)| \leq a_j + c_j.$$

This proves (16.1); no attainment assertion for the original nonfinite minimum is required. The visible comparison is joint with the same selector, not its hidden marginal.

Given a target width $\eta > 0$, enumerate presentations until their certified sum of errors is less than $\eta/4$. For that presentation, Corollary 6.3 produces lower and upper endpoints with gap less than $\eta/2$. Retain maxima of all verified lower endpoints and minima of all verified upper endpoints. The resulting nested physical intervals have width less than η and converge. Zero lower bounds are always available, so this argument also covers zero deficiency and does not ask a strict-positivity certificate to prove equality at a boundary.

The cited constructive collision theorem provides effective graph approximation and marked integration; the finite-report and joint-prefix rationalization theorems then give the presentations and their error bounds. In the collective class, (15.2) replaces the unquantified graph approximation by an explicit rate. Its compact cylinder integration and Gaussian tail bounds give every finite marked-prefix probability. The retained finite-presentation theorem rationalizes these joint prefixes without dividing by a small conditional probability. Thus all links in this specialization are supplied either by an explicitly cited retained theorem or by a proof above, not by assuming a rational physical transition oracle. $\square$

16.1. Exponential acquisition values. The distinction between small total variation and accuracy at an exponential scale is important for the nonlinear-observable direction of the program. The following estimate gives its precise price. It is a theorem about bounded marked acquisition functionals, not a construction of a kinetic large-deviation action.

Theorem 16.2 (Sharp exponential transfer). *Let P, Q be two laws of the same marked transcript with $\|P - Q\|_{TV} \leq \varepsilon$. If h is measurable and $A \leq h \leq B$, then for every $\beta > 0$,*

$$\left| \frac{1}{\beta} \log \mathbb{E}_P e^{\beta h} - \frac{1}{\beta} \log \mathbb{E}_Q e^{\beta h} \right| \leq \frac{1}{\beta} \log \{1 + \varepsilon(e^{\beta(B-A)} - 1)\}. \quad (16.2)$$

The bound is sharp. If a pair of stateless marked presentation channels has error ε uniformly over a declared resource signature, the payoff is evaluated on the same compared marked output interface. Under this convention the same bound holds between optimal infimum values, and between the optimal supremum values, over that signature. The policies are transported through the channels at unchanged budgets.

For the effective physical class of Theorem 16.1, let $\beta_n > 0$ and an upper bound $M_n \geq B_n - A_n$ be computable. One can construct finite rational presentations with certified errors $\varepsilon_n \leq e^{-\beta_n M_n}/n$. Their optimized normalized logarithmic acquisition values then differ from the physical ones by at most $\log(1 + 1/n)/\beta_n$.

Proof. For any bounded g , integration of its level sets gives $|\mathbb{E}_P g - \mathbb{E}_Q g| \leq \varepsilon(\sup g - \inf g)$. Take $g = e^{\beta(h-A)} \in [1, e^{\beta(B-A)}]$. Both expectations are at least one. Their larger-to-smaller ratio is at most $1 + \varepsilon(e^{\beta(B-A)} - 1)$, which proves the estimate. Equality holds when Q is concentrated where $h = A$ and P moves mass ε to a point where $h = B$.

For any admissible physical policy transport it to the presentation and apply the uniform inequality; this gives one inequality between optimal values. Transport in the other direction for the reverse inequality. Approximate optimizing policies suffice, so attainment is unnecessary. The same reasoning works for infima and suprema, but does not replace a common policy by separate phasewise optimizers.

Finally choose a strictly positive rational tolerance below $e^{-\beta_n M_n}/n$, using computable exponential enclosures. The terminating presentation search of Theorem 16.1 attains that tolerance.

Substitution gives the final claim. This is an accuracy-versus-scale construction; it asserts neither an efficient complexity rate nor a large-deviation limit of the microscopic model. $\square$

Remark 16.3 (The downstream implication). Theorems 15.1–16.2 provide a verified fixed-particle link from a nonconserved collision-domain observable to finite-resource marked and exponential acquisition values, with rigorously computable errors. They do not assert particle-number-uniform bounds, a Boltzmann–Grad limit, a nonlinear BBGKY graph corrector on action sublevels, or a range theorem for the historical kinetic Hamiltonian. Those are distinct analytic assertions in the retained program. In particular (16.2) explains why plain unscaled total variation alone does not supply exponential-scale control. All eleven historical components remain in the dependency graph; neither component order nor the existence of an adapter is used as a proof of a missing kinetic theorem.

APPENDIX A. THEOREM DEPENDENCIES AND INFORMATION CONVENTIONS

The comparison conventions of Section 2 supply a compact original behavior image for each fixed signature. Theorem 3.1 uses only that image and its affine tests. Lemma 3.2 and the binomial estimate then give Theorem 3.3. The strict row-cube verifier of Theorem 6.1 is a separate route for checking lower bounds. Proposition 3.6 combines finite strict certificates, executable upper tables and stateless presentation errors; it does not require an inverse constrained randomization theorem.

The collision path has a second, independent finite calculation: Theorem 4.1 solves its marked gated reference experiment, and Lemma 4.2 proves the physical separation of incoming and outgoing detector values. The stateless marked variation estimate joins these two arguments in Theorem 4.3. Proposition 4.4 verifies the effective presentation input using a separated collision root. None of these steps assumes the collective benchmark’s conserved-mode formula.

For optional projections, Bayes’ formula and maximal inequalities first prove Theorem 5.1. Conditional Gaussian change of measure then verifies its hypotheses in Theorem 5.3. The same-latent maximal coupling is used only for process convergence. Finite-register risk transport is instead a data-processing consequence of the original marked variation estimate.

The graph-domain, residual and finite-prefix sections supply the broader physical presentation machinery. Their input is an actual flow, a specified Hilbert realization, a prepared law and an observable with the declared approximation properties. The bound in Theorem 13.1 already expresses the same-budget transfer principle. Theorem 16.1 retains it with the strict polynomial lower module. The present adaptive testing and scattering results add independent-test size control and a collision-generated resource separation; they do not reclassify the transfer principle as a new result each time a certificate module changes.

Throughout the article fresh private randomization has no memory. A hidden persistent selector and a visible persistent selector are separate signatures. A visible comparison is joint with its independent target selector; marginalizing it changes the problem. The computational chart, the revealed-candidate testing game and the latent-preserving convergence coupling are proof constructions, not unpriced channels available to the physical decision maker.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.

- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.